



The Handprint:

a complementary measure of the positive sustainability impact of products

Final Report

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Final content report

Joint report of the four project partners

Project management

Collaborating Centre on Sustainable Consumption and Production (CSCP) gGmbH

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Taking the Handprint approach further – collaboration between CSCP & ELFIN Consulting GmbH

We are pleased to make the final report available in English, expanding its reach to a wider international audience. Our decision to translate the report reflects our dedication to fostering an inclusive and collaborative environment conducive to the exchange of ideas and the advancement of innovations.

It is important to acknowledge that this effort represents just the initial stage, setting the foundation for further development of the Handprint approach. Over the years that followed the closure of the Handprint project, CSCP has been constantly contacted by organizations, companies, universities, associations and many other actors that were interested in exploring the potential of a practical sustainability assessment in different contexts.

Through the collaboration of CSCP and ELFIN Consulting, combining our distinct expertise, we aim to provide a practical and actionable framework to enhance the positive impact of these stakeholders, building upon the great research work done by the project partners of Handprint project funded by the BMBF.

CSCP – who we are & what we do



The Collaborating Centre on Sustainable Consumption and Production (CSCP) is an international non-profit Think and Do tank that works with businesses, policy makers, partner organisations and civil society towards a good life. We have been catalysing change since 2005 through a collaborative and inclusive approach. We strive to empower people, political institutions and organisations to achieve tangible sustainable impacts with positive global ripple effects. Together with key stakeholders we want to mainstream sustainable practices and lifestyles.

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[ELFIN Consulting](https://elfin-consulting.com/)¹ is a business consultancy showcasing over a decade of experience in guiding clients through the intricacies of bridging strategy and implementation. ELFIN's comprehensive range of services encompasses marketing and communications, agile project management, new business development, organizational change, digital transformation, and stewardship – all with a keen focus on sustainability, along with stakeholder strategies. At ELFIN Consulting, we're deeply committed to collaborating with the people driving sustainable business initiatives.

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1 The Handprint Project

1.1 Background

The Handprint project was funded by the Federal Ministry of Education and Research (BMBF) as part of the "Research for Sustainable Development" program (in German: "Forschung für Nachhaltigkeit (FONA)") within the funding priority area of "Socio-ecological Research". This overarching program is aimed, among other things, at implementing findings from sustainability research into innovative applications, products, or services in companies, with the goal of making societal actions more sustainable (BMBF, 2016). In doing so, the BMBF contributes to a "Sustainable Economy," characterized by being both competitive and compatible with the environment and social justice (BMBF, 2013). The Handprint project plays a fundamental role in realizing this vision by developing a methodological approach based on scientific principles. This approach seeks to make the ecological, economic, and social sustainability impacts of products measurable and assessable, thereby providing a foundation for communication.

The existing approach of the ecological Footprint primarily focuses on the negative ecological impacts of individuals, organizations, or countries. In contrast, the Handprint captures the societal added value or positive sustainability effects of products and incorporates the economic and social dimensions into consideration. The Handprint is thus understood as a complementary measure to the ecological Footprint. For a comprehensive assessment of sustainability effects, it is recommended to capture not only the positive effects of the Handprint but also the negative ecological, economic, and social impacts. The result of this assessment, the Handprint of a product, represents a new theoretical and innovatively applicable approach that can contribute to aligning economy, society, and the environment in the long term.

Within this initiative, the Handprint was initially conceptualized and subsequently tested in pilot projects using products from three different sectors as examples. Finally, the approach was publicized with the aim of scaling it up.

1.2 Scientific starting point

For both science and practice, the sustainability assessment of organizations and products is gaining increasing importance, surpassing conventional financial performance measurement, and aiming for a more complex consideration of performance indicators that reflect all aspects of economic, ecological, and social value creation (DeNisi & Smith, 2014; Maas, Schaltegger & Crutzen, 2016b; Richard, Devinney, Yip & Johnson, 2009; Schaltegger & Burritt, 2012). Current sustainability-related performance evaluation primarily focuses on identifying negative sustainability impacts (e.g.,

accidents and fatalities, carbon dioxide emissions, or total operating costs of products; Schaltegger & Csutora, 2012). Largely overlooked, however, is sustainable value creation, i.e., the positive sustainability-related impacts of products or business activities. This neglects important aspects of sustainability performance, missing the opportunity to drive real progress toward sustainability in companies and society by evaluating positive value creation (Beske-Janssen, Johnson & Schaltegger, 2015).

Despite numerous recent calls for a more intensive examination of the assessment of positive sustainability performance (Antolín-López, Delgado-Ceballos & Montiel, 2016; Beske-Janssen et al., 2015; Delmas, Etzion & Nairn-Birch, 2013; Ekener, Hansson & Gustavsson, 2016; Pauw, Kandachar & Karana, 2014; Sala, Farioli & Zamagni, 2013; Schaltegger & Burritt, 2012), only a few researchers have dedicated themselves to this topic (Wilhelm, Hutchins, Mars & Benoit-Norris, 2015). Kroeger and Weber (2015) argue that there is a lack of assessment approaches with general, standardized measures that clearly reflect positive sustainability performance. Without such general and overarching measures, consistent and unambiguous results cannot be achieved, forming the basis for decisions and actions towards sustainable value creation (Maas, Schaltegger & Crutzen, 2016a; Schaltegger & Burritt, 2010).

The quest for general sustainability indicators has often been guided by how the status quo can pragmatically be measured and assessed rather than by what should be evaluated (Salvado, Azevedo, Matias & Ferreira, 2015). As a result, economic and ecological indicators still dominate the field of sustainability assessment, while social indicators only became the subject of discussion later (Beske-Janssen et al., 2015). Furthermore, ecological assessment approaches (e.g., life cycle assessments; LCA) primarily focus on negative impacts and effects that can be quantitatively well measured. More challenging ecological phenomena, such as biodiversity, are often not considered or only considered through highly simplified, quantitatively easy-to-measure metrics (such as land use). In contrast, it is often argued that positive effects arise, on one hand, from the reduction of environmental problems and, on the other hand, particularly from the social dimension of sustainability (e.g., Ekener et al., 2016; Vinyes, Oliver-Solà, Ugaya, Rieradevall & Gasol, 2013), as the ultimate purpose of social measures is to improve the well-being of people (Kroeger & Weber, 2015). Although social assessment approaches like Social Life Cycle Assessment (SLCA) have the potential to capture both negative and positive effects, they are still in the development stage and face various challenges, making them not yet widely accepted and applied (Ekener et al., 2016).

Criticism has been directed towards the limited significant progress in the methodological development of Social Life Cycle Assessments (SLCAs) in recent years, resulting in a standstill in their implementation in corporate practice (Martínez-Blanco, Lehmann, Chang & Finkbeiner, 2015). For instance, Baumann, Arvidsson, Tong, and Wang (2013) express skepticism regarding the SLCA Guidelines and Methodological Sheets of the Life Cycle Initiative by the United Nations Environment

Program (UNEP) and the Society of Environmental Toxicology and Chemistry (SETAC) (UNEP & SETAC, 2009, 2013). They criticize the lack of empirical foundation and systematization in the SLCA Guidelines, highlighting that many indicators, such as child labour, are ideologically biased and may be ambiguous in different cultural contexts (Baumann et al., 2013).

Arcese, Lucchetti, Massa, and Valente (2016) conclude that SLCA has become a highly fragmented field over time, with none of the numerous approaches dominating, resulting in no consensus on relevant (negative and positive) evaluation categories and indicators. Thus, neither social nor ecological and/or economic approaches currently systematically exploit their potential for assessing positive effects. Lastly, the connections, especially between ecological and social aspects, as well as between social and economic aspects, are not explicitly considered by most methods (Beske-Janssen et al., 2015). Partially addressed are connections between ecological and economic aspects, such as cost and emission savings (Kabongo & Boiral, 2017), but even this is often a secondary consideration beyond the actual measurement.

The research project "The Handprint: A complementary measure of positive sustainability impact of products" aimed to develop and test an empirically grounded method for assessing positive sustainability aspects. The method is intended to identify approaches for creating positive ecological, economic, and social sustainability impacts of products, thereby promoting more sustainability in companies. The central question guiding the method development was: How can positive sustainability impacts of products be assessed?

Further details, including additional information found in Working Papers 1 and 2 of the Handprint project can be accessed at www.handabdruck.org.

1.3 Project structure and participants

The project took place from June 1, 2015, to May 31, 2018, spanning a duration of 36 months, with a budget allocation of €935,273. The initiative was structured into three work packages, and a project management team was established for coordination.

The first work package focused on developing a method for scientifically analysing and evaluating the Handprint of a product. This method was then tested in the second work package through selected case studies involving practical partners (SMEs) from various industries. Leveraging a network established during the project, the insights gained were disseminated to other companies and societal stakeholders. Throughout the entire project, the work packages were consistently interconnected to foster synergies.

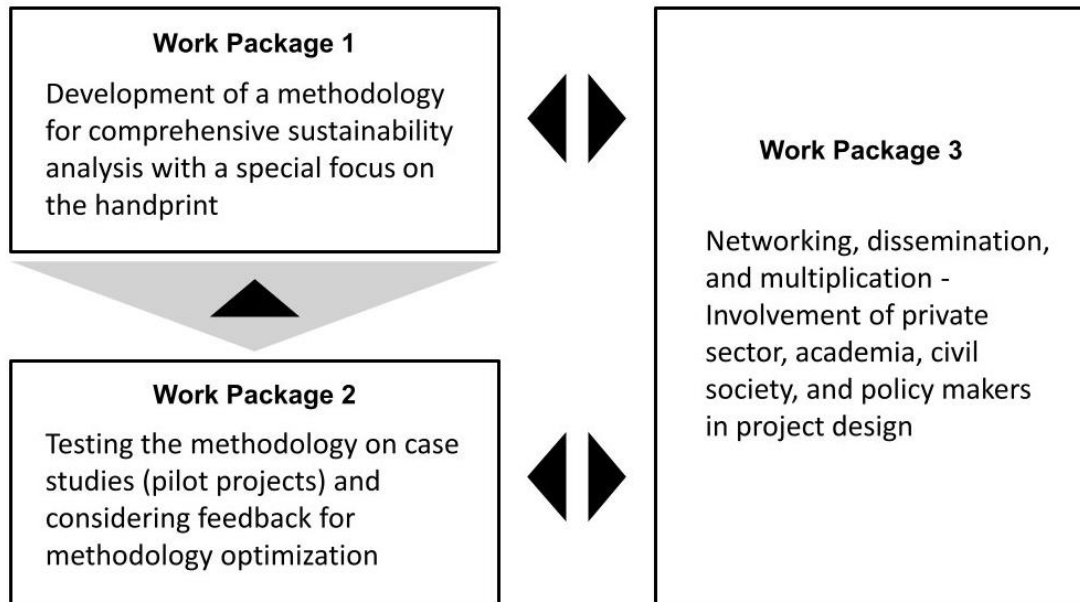


Figure 1: Work packages of the "Handprint" research project

The following individuals were involved in the execution of the project:

- Collaborating Centre on Sustainable Consumption and Production (CSCP) gGmbH (Project Management)
 - Janpeter Beckmann
 - Patrik Eisenhauer
 - Christoph Hermann
 - Birte Schnurr
- Center for Sustainable Corporate Leadership (ZNU), University of Witten/Herdecke
 - Dr. Ulrike Eberle
 - Marianne Schmid
 - Dr. Christian Geßner
 - Dr. Imke Schmidt
- Centre for Sustainability Management (CSM), Leuphana University Lüneburg
 - Prof. Dr. Stefan Schaltegger
 - Samantha Luisa Silva
 - Marten Stock
- University of Hohenheim / University of Kassel

- Prof. Dr. Rüdiger Hahn
- Dr. Michael Kühnen

An overview of the work plan, schedule, and milestones is provided in the following diagram:

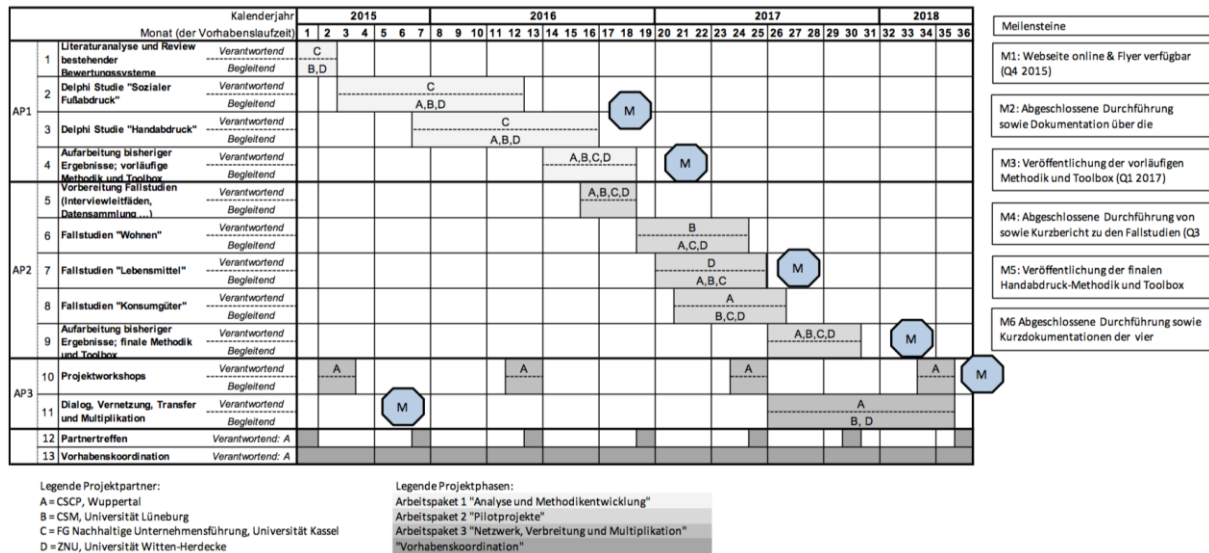


Figure 2: Overview of work and time schedule as well as milestones

In terms of content, the approach follows the multi-method approach. This, along with a detailed description and execution of the work packages, is presented in Chapter 2.

2 Project implementation

2.1 Introduction of challenges and objectives

Despite the undisputed necessity for a more in-depth exploration of the assessment of positive sustainability impacts (Beske-Janssen, Johnson & Schaltegger, 2015; Ekener, Hansson & Gustavsson, 2016; Sala, Farioli & Zamagni, 2013), few initiatives have addressed this topic to date (Wilhelm, Hutchins, Mars & Benoit-Norris, 2015). Consequently, at present, neither social nor ecological nor economic approaches systematically tap into their potential for assessing positive effects. The Handprint project aimed to contribute to the transformation of the economy and society toward a more sustainable development by pursuing the following objectives:

- transparently measuring and evaluating positive sustainability impacts associated with products and their production processes regarding their added value for society, the environment, and the economy

- systematically assessing ecological, social, and ideally economic sustainability impacts together
- exploring the feasibility of a cross-industry approach
- providing guidance for designing more sustainable products and fostering more sustainable business practices through the assessment of both positive and negative sustainability impacts
- discussing approaches to use assessment results as a foundation for clearer communication

The project envisioned regular stakeholder involvement, with the overarching goal of gradually establishing a project-accompanying and expanding network comprising academia, businesses, and civil society. This network played and continues to play a pivotal role in making the Handprint method and supplementary project results applicable to additional products and industries, disseminating the Handprint in politics, science, and civil society, and thereby contributing to the overarching project goal of fostering sustainable societal change.

2.2 Research approach and methodological procedure

The complexity of positive sustainability impacts and the intricacies of a holistic assessment posed and continues to pose a central challenge for the development of the method. Consequently, the following research questions were addressed:

- What definitions and methods exist in academic literature and selected practical examples regarding a positive sustainability contribution of products or companies?
- How do the concepts of Ecological Footprint and Handprint relate to each other?
- What sustainability topics, indicators, and evaluation criteria should form the basis of the Handprint?
- How can economic factors be represented as part of the Handprint?
- How can the sustainability contribution of a product be measured and assessed? Which scale is suitable?
- To what extent can positive sustainability impacts be aggregated into a metric, and how well can such a metric capture the complexity of the issue? Can both ecological and social as well as economic aspects be integrated into a single metric?
- What contribution to the sustainability transformation of society can a company make through its products?

To achieve the set goals and manage the complexity of a potential comprehensive (positive and negative) sustainability assessment, the application of multiple methods offered the advantage of

avoiding a one-sided and narrow perspective. The combination of several systematic analyses provided a comprehensive overview of the current state and identified gaps in knowledge and application. These insights formed the basis for the further development of the Handprint method. Subsequent Delphi studies and stakeholder workshops supported the consideration of diverse requirements and criteria in the development, addressing dimensional conflicts in a holistic sustainability perspective. The multi-method approach ensured that a scientifically grounded, yet practical and solution-oriented method was developed, combining a comprehensive and valid sustainability assessment with high applicability.

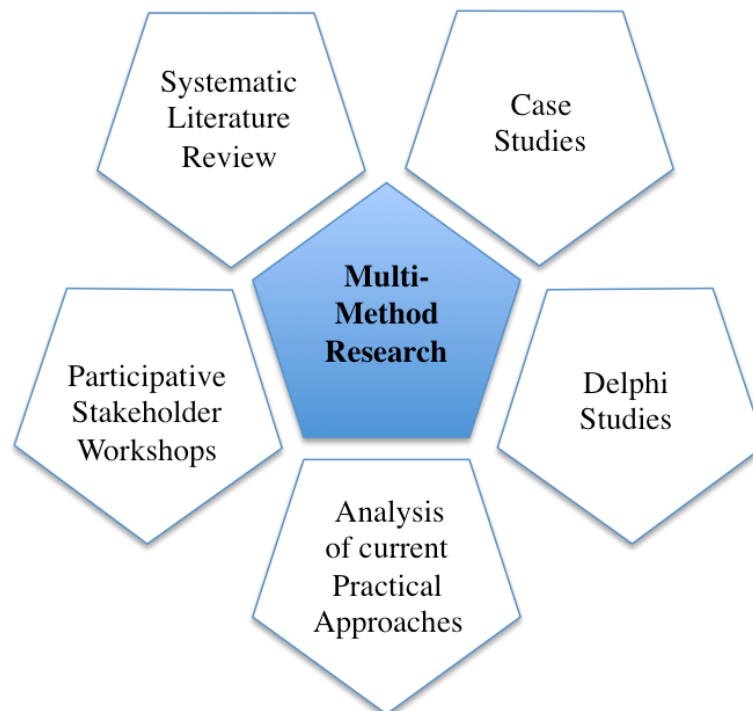


Figure 3: Overview of Applied Research Methods

Following the iterative approach of the research project, the results of various conducted analyses were regularly consolidated and further developed in internal coordination and discussion rounds within the project. The mode of exchange was chosen based on needs and consisted of workshops, teleconferences, and written communication.

For more detailed information, refer to Working Paper 2 of the Handprint project, available at www.handabdruck.org.

2.3 Work package 1: Analysis and method development

The objective of Work Package 1 (WP1) was to conduct a comprehensive inventory of relevant existing assessment methods and measurement approaches for the ecological and social aspects, as well as the negative and positive impacts of products. The already existing approaches served as a basis for the development of the Handprint methodology. Additionally, two Delphi studies were conducted: the first Delphi study focused on the social Footprint, while the second one focused on the Handprint, aiming to create the possibility of a holistic assessment.

Approach:

In the first step, systematic literature analyses were conducted, focusing on selected relevant aspects of the project. This included, among other things, analysing literature on the following topics:

- Sustainability assessment methods for businesses and products
- Definitions of and engagement with "positive sustainability" in sustainability assessment methods
- Possibilities for reducing complexity in various assessment methods
- Aggregation options

For the systematic analysis of existing literature and practice approaches, the research approach of Denyer and Tranfield (2009) and Tranfield, Denyer, and Smart (2003) was utilized.

To consider current developments from practice, in addition to reviewing academic literature, an analysis of published methods for sustainability assessment of products developed by selected companies was conducted. For the systematic analysis of practical approaches, the research approach of Denyer and Tranfield (2009) was also utilized. To find cases from corporate practice, initially, companies listed in the Standard & Poor's 500 (S&P 500) and Stoxx Europe 600 (STXE 00) indices (as of January 30, 2015) were identified. Subsequently, a Google search was conducted by combining various company names with the keywords "sustainability measurement", "sustainability assessment" and "sustainability analysis". With this approach, 22 companies were identified that publicly state their use of a proprietary sustainability assessment approach with social indicators.

Building upon the findings of the systematic analysis of scientific studies and practical examples, two Delphi studies were conducted with experts in the field of Life Cycle Assessments (LCA). The aim was to achieve a comprehensive and coherent understanding of social and positive product sustainability assessment. A Delphi study generally aims to structure a group communication process that allows a group of experts to address a complex problem. Such a study is thus characterized as an anonymous, iterative, multi-stage survey process, where moderators aggregate the group's opinion and provide feedback to individual participants after each round of questioning (Linstone, Turoff & Helmer, 2002). Schmidt (1997) outlines a structured approach to conducting a Delphi study.

In the Handprint project, this structure was adapted for two Delphi studies (on the assessment of social and positive sustainability of products) drawing from the same pool of experts.

This allowed for a cohesive and shared understanding of the key points in assessing social and positive sustainability.

During the project's initial phase and parallel to the first Delphi study, a kick-off workshop with stakeholders was conducted to introduce the Handprint project to an interested professional audience and lay the foundation for a project-accompanying and growing network of science, business, and civil society (CSCP, CSM, University of Hohenheim & ZNU, 2015). Discussions with stakeholders revealed how companies - consciously or unconsciously - deal with positive sustainability impacts and their assessment. Furthermore, requirements and expectations regarding the Handprint methodology and the project were discussed. After the completion of the Delphi studies, a second stakeholder workshop followed to present the preliminary results and the Handprint methodology for critical reflection by stakeholders (CSCP, CSM, University of Hohenheim & ZNU, 2016). Stakeholders discussed previously selected potential assessment categories and questions regarding an appropriate number of categories and the practicability of the Handprint methodology in parallel working groups. It was explicitly taken into consideration whether the practicability of the Handprint methodology could be improved by reducing the number of assessment categories, using a set of mandatory and product-independent "must categories" that could be supplemented by product-dependent "optional categories".

Results:

2.3.1 SLCA and social indicators in scientific literature

The first identified articles addressing LCA and social indicators date back to the year 2003. This relatively late entry into the topic is not surprising, considering that the first articles on the general SLCA concept were published in the mid-1990s, and the discussion about specific life cycle-oriented social indicators intensified only at the beginning of the new millennium, with some time delay. After an initial increase in publications, there was a decline in 2007 due to the initial challenges to integrate social indicators into the LCA framework and the unclear concept of "integration." Only in recent years there has been a noticeable increase in publications, driven by the release of guidelines and frameworks for the social responsibility of organizations (from which social indicators are derived).

In terms of the distribution of research approaches, non-empirical (conceptual) articles constitute approximately 42% of the analysed literature. Among these, literature reviews (eight articles) only partially provide insights into social indicators as they focus on overarching aspects such as the development of models for sustainable supply chain management (e.g., Brandenburg, Govindan,

Sarkis & Seuring, 2014; Seuring, 2013). Other conceptual articles (33) either identify or develop social indicators and discuss their integration into a holistic life cycle-oriented sustainability assessment. Empirical articles make up 58% of the analysed literature. The relatively high number of 41 quantitative studies (~43% of the literature) indicates the emerging maturation of social sustainability assessment, progressing from a conceptual foundation to the quantitative application of social indicators. However, this development may be open to scrutiny considering the comparatively low number of 15 qualitative studies (~15% of the literature), as social aspects are often qualitative and not always easily quantifiable. Therefore, the completeness of social indicators might be questioned if qualitative indicators are neglected in favor of easily quantifiable aspects.

2.3.2 SLCA and social indicators in business practice

Similar to scientific literature, the integration of social indicators into corporate practices began in the early 2000s and has intensified since then. When comparing research literature and corporate practices, it is noticeable that in the literature, social indicators are often formulated abstractly (e.g., child labor) without clearly specifying whether they have a negative (e.g., physical, and mental burdens on working children) or positive (e.g., financial contribution of working children to family income and survival) impact. Only a small proportion of the analysed literature uses indicators clearly stating a positive impact. In contrast, most of the identified assessment approaches from corporate practices address positive social indicators. Overall, the indicators in both scientific literature and identified corporate approaches are often formulated in such a generic manner that it is not clearly discernible whether they are evaluated through qualitative descriptions or quantitative measurements.

2.3.3 Approaches to positive sustainability measurement in the scientific literature

To achieve sustainable development, positive sustainability contributions that go beyond the reduction of harmful effects are increasingly gaining importance. A crucial prerequisite for implementing this at the corporate level is the availability of appropriate methods for determining, measuring, and assessing sustainability impacts. A systematic literature analysis was conducted to assess the extent to which existing approaches to sustainability determination, measurement, and evaluation already capture positive sustainability contributions. The results from the analysis of the 318 identified articles indicate that, while the general relevance of positive sustainability effects is widely discussed in the scientific literature, this is only partially reflected in existing methods. These methods largely focus on negative impacts, while positive contributions are currently only implicitly considered. This is due to the fact that a majority of approaches rely on quantitative measures, which

are more objectively measurable and usable, but may only inadequately or insufficiently capture positive effects. Consequently, there is a need to develop existing approaches further or create new instruments for positive sustainability assessment to meet the requirements of a truly comprehensive sustainability evaluation, thereby providing an improved, integrated decision-making basis for businesses.

2.3.4 Results of the Delphi studies on social and positive sustainability assessment

The results of the analysis of existing literature and practical approaches indicate an increasing relevance of social and positive sustainability effects, but also show that the assessment of these effects lack consistency and are underdeveloped. Therefore, these results were used to prepare two exploratory Delphi studies, which represent an initial step toward achieving consensus and prioritizing the most important assessment categories of social and positive sustainability throughout the product lifecycle. The outcome of these studies was a ranking of the most crucial aspects, which should be incorporated into the preliminary Handprint method as mandatory and product-independent assessment categories.

2.3.5 Developed framework method

Utilizing a multi-method research approach, an initial framework was developed. The Handprint methodology follows the approach of the Life Cycle Assessment (LCA) methodology (ISO 14040/44). To evaluate and identify crucial sustainability aspects, the methodology leverages the Sustainable Development Goals (SDGs) outlined by the United Nations (United Nations, 2015). Sustainability impacts are measured through indicators, with the SDGs serving as the benchmark for assessment.

The Handprint methodology follows the approach of comparative product Life Cycle Assessment (ISO 14040/44), which is widely acknowledged and scientifically grounded for product evaluation. Additionally, many of the analysed approaches for sustainability assessment of products, such as Product Sustainability Assessment (PROSA; Grießhammer et al., 2007) and SEEbalance (Schmidt et al., 2004), are based on the Life Cycle Assessment methodology. Furthermore, Life Cycle Assessment is an accepted instrument in many companies for capturing environmental impacts of products and production processes, aiming to reduce negative effects. Aligning the Handprint methodology with the Life Cycle Assessment provides the advantage of easier compatibility. Consequently, the Handprint methodology involves an analysis of the life cycle ("from cradle to grave") of the considered systems, focusing on a clearly defined functional unit. In alignment with the Life Cycle Assessment methodology (Klöpffer & Grahl, 2009), the goal and scope are defined, a material flow analysis is conducted, and an impact assessment is performed. Unlike the Life Cycle Assessment methodology,

there is an additional step in the Handprint methodology: the results of the impact assessment are evaluated with regards to their sustainability contributions with respect to the SDGs. Subsequently, similar to the approach of a Life Cycle Assessment, the results are analysed and interpreted. The determination of measurable indicators was conducted through an iterative process. Initially, an analysis was performed to identify which indicators in the ecological, social, and economic sustainability dimensions are already in use or should be utilized according to stakeholders' perspectives. To achieve this, a systematic literature analysis and a two-stage Delphi study were carried out as the first steps to achieve an initial selection and prioritization of key social and positive sustainability indicators along the life cycle of products. Following the establishment of the "Sustainable Development Goals (SDGs)" as the benchmark for assessment, an analysis was conducted to identify the information needed to make the sustainability of products measurable, as defined by the SDGs. For this purpose, the 17 main goals and all 169 sub-goals were analysed. During this analysis, it was determined that not all SDGs represent goals to which companies can contribute through their offered products or business activities. Therefore, indicators are only proposed for those SDGs to which companies can contribute through their products or business activities. The goal is to create a broad pool of indicators, which will be tested in case studies in the next step. The indicators are assigned to five areas:

- social aspects
- human health
- environmental aspects
- economic aspects
- governance

The following Table 1 provides an overview of all indicators comprising the indicator pool.

Table 1: Overview of indicators

Dimension	Indicators	SDG #	Sustainable Development Goal
Social aspects	Average wage of workers compared to minimum wage and minimum livelihood	1.1	By 2030, eradicate extreme poverty for all people everywhere, currently measured as people living on less than \$1.25 a day.
		8.5	By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.

Dimension	Indicators	SDG #	Sustainable Development Goal
	Expenditure on social security	1.3	Implement nationally appropriate social protection systems and measures for all, including floors, and by 2030 achieve substantial coverage of the poor and the vulnerable.
		8.5	By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.
	Ratio of men's to women's pay (at different salary levels)	8.5	By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.
	Number of cases of child labour along the product life cycle	8.7	Take immediate and effective measures to eradicate forced labour, end modern slavery and human trafficking and secure the prohibition and elimination of the worst forms of child labour, including recruitment and use of child soldiers, and by 2025 end child labour in all its forms.
	Promoting diversity and equal treatment	8.5	By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.
Health (risk prevention)	Number and/or time lost due to occupational accidents/illnesses	3.9	By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination.
	Number of workers with access to protective clothing	8.8	Protect labour rights and promote safe and secure working environments for all workers, including migrant workers, in particular women migrants, and those in precarious employment.
	Human toxicity potential Ozone depletion potential Summer smog potential Radiation potential Particulate matter potential	3.9	By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination.
		12.4	By 2020, achieve the environmentally sound management of chemicals and all wastes throughout their life cycle, in accordance with agreed international frameworks, and significantly reduce their release to air, water and soil in order to minimize their adverse impacts on human health and the environment.

Dimension	Indicators	SDG #	Sustainable Development Goal
Environment	Terrestrial biodiversity potential	2.4	By 2030, ensure sustainable food production systems and implement resilient agricultural practices that increase productivity and production, that help maintain ecosystems, that strengthen capacity for adaptation to climate change, extreme weather, drought, flooding and other disasters and that progressively improve land and soil quality.
		2.5	By 2020, maintain the genetic diversity of seeds, cultivated plants and farmed and domesticated animals and their related wild species, including through soundly managed and diversified seed and plant banks at the national, regional and international levels, and promote access to and fair and equitable sharing of benefits arising from the utilization of genetic resources and associated traditional knowledge, as internationally agreed.
		6.6	By 2020, protect and restore water-related ecosystems, including mountains, forests, wetlands, rivers, aquifers and lakes.
		12.4	By 2020, achieve the environmentally sound management of chemicals and all wastes throughout their life cycle, in accordance with agreed international frameworks, and significantly reduce their release to air, water and soil in order to minimize their adverse impacts on human health and the environment.
		15.1	By 2020, ensure the conservation, restoration and sustainable use of terrestrial and inland freshwater ecosystems and their services, in particular forests, wetlands, mountains and drylands, in line with obligations under international agreements.
		15.4	By 2030, ensure the conservation of mountain ecosystems, including their biodiversity, in order to enhance their capacity to provide benefits that are essential for sustainable development.
		15.5	Take urgent and significant action to reduce the degradation of natural habitats, halt the loss of biodiversity and, by 2020, protect and prevent the extinction of threatened species.

Dimension	Indicators	SDG #	Sustainable Development Goal
	Wastewater volumes Freshwater eutrophication potential Freshwater toxicity potential	6.3	By 2030, improve water quality by reducing pollution, eliminating dumping and minimizing release of hazardous chemicals and materials, halving the proportion of untreated wastewater and substantially increasing recycling and safe reuse globally.
	Water volumes used Water scarcity	6.4	By 2030, substantially increase water-use efficiency across all sectors and ensure sustainable withdrawals and supply of freshwater to address water scarcity and substantially reduce the number of people suffering from water scarcity.
	Land use	6.6	By 2020, protect and restore water-related ecosystems, including mountains, forests, wetlands, rivers, aquifers and lakes.
	Use of resources (renewable energies, energy efficiency, resource efficiency, resource consumption)	7.2	By 2030, increase substantially the share of renewable energy in the global energy mix.
		7.3	By 2030, double the global rate of improvement in energy efficiency.
		8.4	Improve progressively, through 2030, global resource efficiency in consumption and production and endeavours to decouple economic growth from environmental degradation, in accordance with the 10-year framework of programs on sustainable consumption and production, with developed countries taking the lead.
		12.2	By 2030, achieve sustainable management and efficient use of natural resources.
		12.3	By 2030, halve per capita global food waste at the retail and consumer levels and reduce food losses along production and supply chains, including post-harvest losses.
	Waste quantities	12.4	By 2020, achieve the environmentally sound management of chemicals and all wastes throughout their life cycle, in accordance with agreed international frameworks, and significantly reduce their release to air, water and soil in order to minimize their adverse impacts on human health and the environment.

Dimension	Indicators	SDG #	Sustainable Development Goal
		12.5	By 2030, substantially reduce waste generation through prevention, reduction, recycling and reuse.
	Ecotoxicity potential	6.3	By 2030, improve water quality by reducing pollution, eliminating dumping and minimizing release of hazardous chemicals and materials, halving the proportion of untreated wastewater and substantially increasing recycling and safe reuse globally.
		12.4	By 2020, achieve the environmentally sound management of chemicals and all wastes throughout their life cycle, in accordance with agreed international frameworks, and significantly reduce their release to air, water and soil in order to minimize their adverse impacts on human health and the environment.
	Global warming potential	13	Take urgent action to combat climate change and its impacts.
	Marine eutrophication potential Marine toxicity potential	14.1	By 2025, prevent and significantly reduce marine pollution of all kinds, from land-based activities, including marine debris and nutrient pollution.
	Marine acidification potential	14.3	Minimize and address the impacts of ocean acidification, including through enhanced scientific cooperation at all levels.
	Marine biodiversity potential	14.4	By 2020, effectively regulate harvesting and end overfishing, illegal, unreported and unregulated fishing and destructive fishing practices and implement science-based management plans, in order to restore fish stocks in the shortest time feasible, at least to levels that can produce maximum sustainable yield as determined by their biological characteristics.
		14.5	By 2020, conserve at least 10 per cent of coastal and marine areas, consistent with national and international law and based on the best available scientific information.
	Soil quality potential	15.3	By 2030, combat desertification, restore degraded land and soil, including land affected by desertification, drought and floods, and strive to achieve a land degradation-neutral world.

Dimension	Indicators	SDG #	Sustainable Development Goal
		15.5	Take urgent and significant action to reduce the degradation of natural habitats, halt the loss of biodiversity and, by 2020, protect and prevent the extinction of threatened species.
Economic aspects	Contribution to resource efficiency (efficient use of resources, reuse/recycling content, etc.)	8.4	Improve progressively, through 2030, global resource efficiency in consumption and production and endeavour to decouple economic growth from environmental degradation, in accordance with the 10-year framework of programs on sustainable consumption and production, with developed countries taking the lead.
		12.2	By 2030, achieve sustainable management and efficient use of natural resources.
		12.3	By 2030, halve per capita global food waste at the retail and consumer levels and reduce food losses along production and supply chains, including post-harvest losses.
		12.5	By 2030, substantially reduce waste generation through prevention, reduction, recycling and reuse.
	Dissemination of product-related technological sustainability solutions	17.6	Enhance North-South, South-South and triangular regional and international cooperation on and access to science, technology and innovation and enhance knowledge sharing on mutually agreed terms, including through improved coordination among existing mechanisms, in particular at the United Nations level, and through a global technology facilitation mechanism.
		17.7	Promote the development, transfer, dissemination and diffusion of environmentally sound technologies to developing countries on favourable terms, including on concessional and preferential terms, as mutually agreed.
	Contribution to reducing waste	12.4	By 2020, achieve the environmentally sound management of chemicals and all wastes throughout their life cycle, in accordance with agreed international frameworks, and significantly reduce their release to air, water and soil in order to minimize their adverse impacts on human health and the environment.
		12.5	By 2030, substantially reduce waste generation through prevention, reduction, recycling and reuse.

Dimension	Indicators	SDG #	Sustainable Development Goal
Governance	Capacity building - creating new or increasing existing capacities	4.7	By 2030, ensure that all learners acquire the knowledge and skills needed to promote sustainable development, including, among others, through education for sustainable development and sustainable lifestyles, human rights, gender equality, promotion of a culture of peace and non-violence, global citizenship and appreciation of cultural diversity and of culture's contribution to sustainable development.
		13.3	Improve education, awareness-raising and human and institutional capacity on climate change mitigation, adaptation, impact reduction and early warning.
	Investments in R&D with a focus on sustainability	9.5	Enhance scientific research, upgrade the technological capabilities of industrial sectors in all countries, in particular developing countries, including, by 2030, encouraging innovation and substantially increasing the number of research and development workers per 1 million people and public and private research and development spending.
		17.3	Mobilize additional financial resources for developing countries from multiple sources.
		17.7	Promote the development, transfer, dissemination and diffusion of environmentally sound technologies to developing countries on favourable terms, including on concessional and preferential terms, as mutually agreed.
		17.16	Enhance the Global Partnership for Sustainable Development, complemented by multi-stakeholder partnerships that mobilize and share knowledge, expertise, technology, and financial resources, to support the achievement of the Sustainable Development Goals in all countries, in particular developing countries.
	Sustainability (risk) management in the company and in the supply chain	12.2	By 2030, achieve sustainable management and efficient use of natural resources.
		12.6	Encourage companies, especially large and transnational companies, to adopt sustainable practices and to integrate sustainability information into their reporting cycle.

Dimension	Indicators	SDG #	Sustainable Development Goal
		16.3	Promote the rule of law at the national and international levels and ensure equal access to justice for all.
	Compliance with sustainability standards	12.6	Encourage companies, especially large and transnational companies, to adopt sustainable practices and to integrate sustainability information into their reporting cycle.
		12.8	By 2030, ensure that people everywhere have the relevant information and awareness for sustainable development and lifestyles in harmony with nature
		16.6	Develop effective, accountable and transparent institutions at all levels
	Integration of sustainability information into reporting	12.6	Encourage companies, especially large and transnational companies, to adopt sustainable practices and to integrate sustainability information into their reporting cycle.
		12.8	By 2030, ensure that people everywhere have the relevant information and awareness for sustainable development and lifestyles in harmony with nature
	Active communication of sustainability aspects to users	12.8	By 2030, ensure that people everywhere have the relevant information and awareness for sustainable development and lifestyles in harmony with nature
	Violations of the law (anti-competitive behaviour, tax violations, environmental laws and regulations, social and labour (protection) laws)	16.3	Promote the rule of law at the national and international levels and ensure equal access to justice for all
	Commitment to the development of sustainability standards	16.6	Develop effective, accountable and transparent institutions at all levels.

2.3.6 Conclusion

The objectives of AP1 were fully achieved, and the results were extensively published, making them widely accessible to the public. Based on the findings from the literature analysis, practical approaches, and Delphi studies, specific indicators for the assessment categories were developed in

internal project discussions. This effort aimed to evolve the Handprint method as part of a comprehensive sustainability analysis. Additionally, the following (preliminary) definition of a comprehensive sustainability analysis was established: "A comprehensive sustainability analysis of products considers both positive and negative ecological, economic, and social impacts throughout the entire product life cycle. In addition to the existing Footprint, which focuses on negative effects, the Handprint, as a complementary measure, captures the positive effects." By integrating positive aspects, the Handprint method can ultimately contribute to a more holistic assessment of the sustainability impacts of products.

2.4 Work package 2: Pilot projects

In WP2, the following questions were examined through case studies:

- How well does the methodology developed in Work Package 1 perform in practice? Where is adaptation needed for practical application?
- Which sustainability impacts or indicators are particularly relevant for investigation? Which indicators should be mandatory, and which ones are optional? Does the relevance of indicators vary across the studied industries?
- What is the added value of the analysis for SMEs? How do companies specifically benefit from calculating the Handprint, and what incentives exist for companies to participate?
- What insights arise from practical implementation? What are the challenges and opportunities for improvement?

The methodology developed in WP1 was tested in case studies using product examples from three different industries. The three project partners, ZNU, CSCP, and CSM, collaborated with a company as a case study partner in each case. Together with these partners, a product was selected as the subject of investigation for the case studies:

- Food industry, company partner Ornu Deutschland GmbH, product example: Kerrygold Butter (Food case studies), conducted by ZNU.
- Consumer goods industry, company partner Nager IT e.V., product example: fair computer mouse (Consumer Goods case studies), conducted by CSCP.
- Construction industry, company partner Werkhaus GmbH, product example: VW T1 Office Organizer (Building & Living case studies), conducted by CSM.

The following case studies were selected for the three product examples:

Food case studies:

- **Biodiversity** case study: The aim was to determine whether the selected indicator and the underlying method provide distinct results. For this purpose, milk production on Irish farms for Kerrygold Butter was compared with milk production in the Allgäu region in small-scale structures.
- **Allocation** case study: Based on selected social indicators, another case study examined the impact of allocation based on mass and based on working hours on the final result.
- **Sustainability risk management** case study: The goal here was to examine the weighting within the various thematic areas of the indicator.

It was also checked whether the indicators identified in the project are suitable for the food sector or whether changes and/or additions are needed.

Consumer goods case studies:

- **Material substitution** case study: The study focused on the impact of material substitution on environmental indicators as well as social, economic, and governance indicators. Additional questions aimed to explore to what extent a company's efforts towards more sustainable alternatives can be adequately reflected in the methodology.
- **Diversity and equal treatment** case study: This study examined how the integration of mentally ill and disabled people in the final assembly stage affects the relevant social indicators and associated SDGs. The goal was to answer the question of how well the chosen measure (integration of people with disabilities) can be appropriately reflected in the method, providing helpful information for both companies and consumers, thus offering actionable insights.
- **Supply chain transparency** case study: Since the aspect of transparency is represented by several indicators, the goal of the case study was to evaluate the distinctiveness of the indicators in practice and to test different indicator designs regarding their evaluation.

Building & living case studies:

SDG 12 “Ensure sustainable consumption and production patterns” primarily addresses companies. As this sustainability goal sets high expectations for companies and simultaneously aligns with changes beyond corporate boundaries, operationalizing and implementing it poses a significant challenge. Despite the relevance of SDG 12 for companies, they tend to report more on SDGs 3 (“Good health and well-being”), SDG 8 (“Decent work and economic growth”), and SDG 13 (“Climate action”)². Following confirmation (through consultation with boards and sustainability representatives) from the case study partner Werkhaus that the identified SDG is of utmost relevance to the company, it

² KPMG (2018) [How to report on the SDGs – What it looks like and why it matters.](#)

underwent examination in two separate case studies: one concentrating on environmental aspects and the other on governance.

- Case study SDG 12 – “Ensure sustainable consumption and production patterns” (focus on environmental indicator).
- Case study SDG 12 – “Ensure sustainable consumption and production patterns” (focus on governance indicator).

Based on the analysis of SDG 12, the necessity to consider a transformational impact was derived. Initial conceptual considerations were developed in the case study, regarding the case study partner, and in comparison, to another German SME. Additionally, the case study critically questioned the extent to which the identified positive contributions to sustainable development are genuinely positive and explored how a company could contribute to the sustainability transformation of the market, society, and the environment:

- Case study - transformational impact

The following presents the results of the conducted case studies.

2.4.1 Food case studies

Authors: Dr. Ulrike Eberle, Marianne Schmid

Selection of the case study partner

For the food sector, the selection of a case study partner for conducting the case studies was carried out among the members of the ZNU partner network. The reason for this is the longstanding and good collaboration with these partners on various topics of sustainable development, which generally increases their willingness to participate as partners in a research project. Additionally, building on existing working relationships is known to facilitate the access to necessary data. Therefore, an initial inquiry was made among the companies in the partner network regarding their willingness to participate in the research project. From the companies that expressed their willingness, one producer of a plant-based product and one producer of an animal-based food product were selected. However, due to a change in ownership of the company, the producer of the plant-based food product withdrew their commitment during the project period. Consequently, only one case study partner, Ornu Deutschland GmbH with the product "Kerrygold Butter" continued to participate.

Implementation

The implementation of the case studies followed the classical approach of a Life Cycle Analysis or ecological Footprint assessment. Initially, the life cycle of Kerrygold Butter up to the "gate" at Ornu Deutschland was researched. Subsequently, together with the case study partner,

considerations were made on how to structure the data collection in the most time-efficient manner for the case study partner.

Based on these results and in consultation with Ornua Deutschland GmbH, a decision was made to collect data at three actor levels:

- at the level of Ornua Deutschland (direct contact)
- at the level of the dairy in Ireland (contact through Ornua Deutschland GmbH)
- at the level of dairy cattle farming in Ireland (contact through Ornua Deutschland GmbH)

Specific data collection questionnaires in German and English were created for each of these three actor levels. These questionnaires were aligned with Ornua Deutschland GmbH and were sent out in April 2017. By October, the data return was complete, although some follow-up data collection was necessary.

Considered Product System

The analysed butter is produced exclusively in Ireland from local milk, then transported chilled in large blocks to Germany, where it is portioned and packaged.

In Ireland, there are approximately 16.000 dairy farms with an average size of 56 hectares, maintaining around 70 cows each. About 14.000 farms supply Kerrygold dairies.

On average, cows in Ireland spend 255 days a year on pasture (equivalent to 70% of the year) and receive minimal supplementary feeding during this time. Farmers also produce silage (twice a year) for winter days. Supplementary feeding includes dairy performance feed, typically composed of cereals (wheat, barley), rapeseed and soybean meal, distillers' grains, and citrus pulp. Except for soybean meal and distillers' grains, the feed components originate from the EU.

The primary product of dairy farming is milk, and in addition, a dairy farm always produces calves. Furthermore, after the slaughter of dairy cows, meat is generated. Additionally, manure and dung are produced, usually spread on the farm's own land.

The milk is delivered to the dairy, which produces, among other products, butter. The butter is frozen in large blocks and gradually transported, also to Germany. Here, it is portioned, packaged, and supplied to retailers.

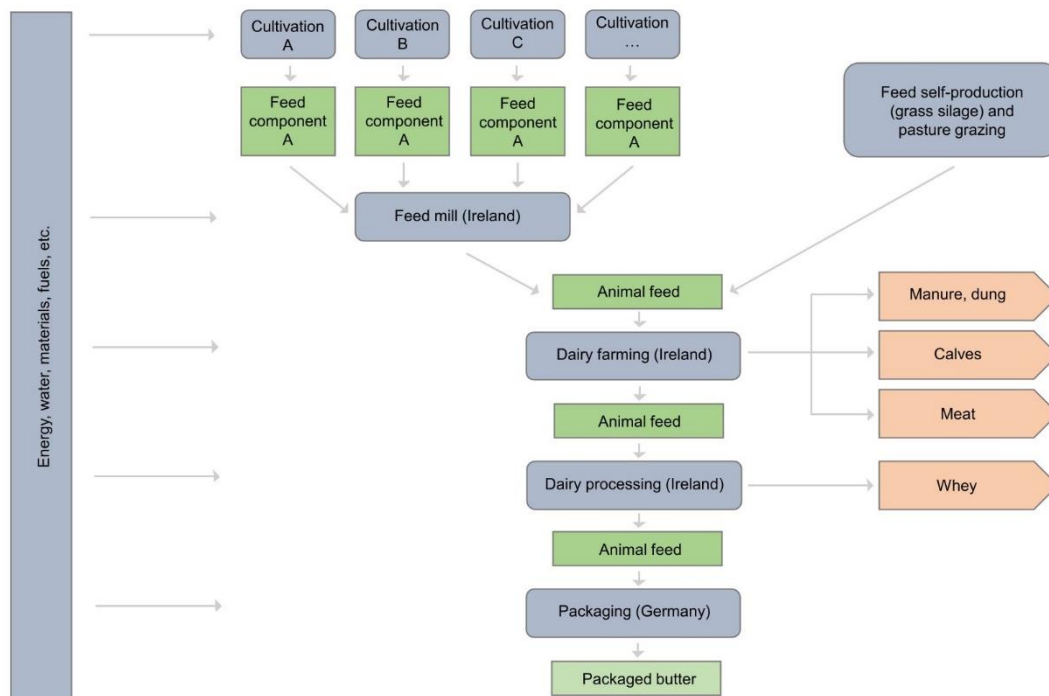


Figure 4: Lifecycle of a Kerrygold butter until before delivery to retail.

Results

The following section presents the results of the food case studies: the first subsection includes the comparison of project indicators with those identified in the food sector. The subsequent sections present the results of the food case studies.

Handprint indicators - food

The aim of the research project "Handprint" was to develop a method that allows the measurement and evaluation of positive sustainability contributions. For this purpose, the Sustainable Development Goals (SDGs) of the United Nations were used as an internationally accepted benchmark for evaluation. Indicators were defined for measurement, which were then applied in various case studies. Changes and additions have emerged in detail from these, partly due to the requirements of the examined product group (food), partly due to necessary adjustments resulting from difficulties in data collection, and partly as clarifications of the original indicator. In Table 2 below, the originally defined potentially positive indicators and the results of the verification in the Food case study are compared. Additional indicators, which are also relevant for sustainability assessment but cannot assume positive values in the evaluation – such as the number of cases of child or forced labour – are not listed in the table.

Table 2: Comparison of the original indicators and the indicators from the food case studies

Project proposal		Suggested changes/additions from the Food case studies	
Indicator	SDG reference	Indicator	SDG reference
Average wage of workers compared to minimum wage and living wage	1.1 8.5	P: Average hourly wage in the lowest income group	1.1
Expenditure on social security	1.3 8.5	C: Coverage of social security	1.3
		Ea: Coverage of sustainability (risk) management ³	2.4, 6.5, 6.6, 12.2, 12.4, 12.5, 13.2, 14.2, 15.1-15.6, 15.9, 16.5, 17.7
		Ef: Number of varieties/ breeds used	2.5
Number of workers with access to protective clothing	3.9 8.8	P	3.9, 8.8, 12.4
		Ea: Number of occupational accidents/time lost due to occupational illnesses	3.9, 8.8, 12.4
		Ea: Proportion of trainees in the workforce (by gender)	4.4
		Ea: Average hours of further education/training per employee (by gender, employee category and age)	4.5
Capacity building (training measures)	4.7 13.3	C: Proportion of employees who have received training on sustainability issues	4.7
Ratio of men's to women's pay (at different salary levels)	8.5	P	5.1, 8.5

³The following topics are addressed in the SDGs: Agriculture, Water use & scarcity, Freshwater ecosystems, Natural resource use, Chemical use, Waste, Climate change, Marine biodiversity, Terrestrial biodiversity, Patents on natural resources, Poverty reduction, Corruption prevention, Promotion of environmentally friendly technologies in developing countries.

Project proposal		Suggested changes/additions from the Food case studies	
Promoting diversity and equal treatment	8.5	Ea: Existence of an equal opportunities officer	5.1
		Ea: Proportion of women in management positions at all hierarchical levels	5.5
		Ea: Availability of drinking water at the workplace	6.1
		Ea: Availability of adequate sanitary facilities (separated by gender, lockable)	6.2
		Ea: Existence of adequate wastewater treatment	6.3
Share of renewable energies in energy consumption	7.2	P	P
		Ea: Share of employees under 30 years of age	8.6
		Ea: Proportion of micro-suppliers (especially from developing countries)	9.3
Investments in R&D (with a focus on sustainability)	9.5 (17.3, 17.7, 17.16)	P: Investments in R&D	9.5
		Ea: Composition of the workforce (by age, gender, disabilities, other groups if applicable)	10.2
Proportion of recycled/reused materials (bulk purchase) (contribution to reducing waste/volume of waste)	12.4. 12.5.	X	

Project proposal		Suggested changes/additions from the Food case studies	
Integration of sustainability information into reporting	12.6 12.8	P	12.6
Compliance with sustainability standards	12.6 12.8 16.6	C: Compliance with sustainability standards at own locations	12.8
		C: Compliance with sustainability standards in the supply chain	12.8
Active communication of sustainability aspects to users	12.8	P: Availability of sustainability information for users	12.8
		Ea: Proportion of employees trained in climate protection	13.3
		Ef: <i>Reporting on fishing practices, fish stocks and fishing locations (only relevant for marine products)</i>	14.4
		Ea: Investments in the conservation and sustainable use of biodiversity/ecosystems	15.a
		Ea: Investments in forest conservation, reforestation and management	15.b
		Ea: Percentage of employees trained in corruption prevention	16.5
		Ea: Existence of a corruption prevention clause in employment contracts	16.5
		Ea: Number of reported cases of corruption	16.5

Project proposal		Suggested changes/additions from the Food case studies	
		Ea: Existence of an independent mediator for corruption cases	16.5
Commitment to the development of sustainability standards	16.6	P: Engagement in multi-stakeholder partnerships for sustainable development	17.17
Dissemination of technological sustainability solutions	17.6 17.7	X	
Contribution to the mobilization of financial resources with a focus on sustainability for developing countries	17.6 17.7	X	
		Ea: Proportion of raw materials/products from developing countries	17.11

*Legend **P**: Indicator/ SDG reference matches, if necessary, the designation was specified; **C**: indicator was changed; **Ea**: generally valid indicator was added; **Ef**: food-specific indicator was added; **X**: indicator is of minor relevance/no significance for food*

The application of the indicators in the Food case studies revealed that adjustments to the indicators were necessary in some cases. Reasons for this included the need to add indicators that should also be applied to other products, the requirement for adjustments to general indicators (e.g., practicality, more precise alignment with the respective SDG, specification), or the necessity for adjustments from the perspective of evaluating a food product (specification, addition of an indicator, removal of an indicator).

2.4.1.1 Case study biodiversity

Based on the data from Irish farms and data for milk production from the Allgäu region, a comparative evaluation was conducted regarding the impact of butter's milk production on biodiversity.

Excursus: Method for Determining Biodiversity Impacts

The method defined in the framework methodology for calculating the impacts on biodiversity is based on Michelsen (2007) and was developed by Lindner et al. (in preparation) and Lindner (2016). The biodiversity potential determined in this way describes the impacts of activities on biodiversity, including the diversity of species and ecosystems, as well as genetic diversity. The inclusion of this impact category in life cycle assessments has been discussed for several years. In 2007, the UNEP-SETAC Life Cycle Initiative developed a framework document on "Environmental Impacts of Land Use" (Milà i Canals et al. 2007), in which it is specified to "calculate the impacts of land use based on the differences in land quality before, during, and after a defined use." The used area, the duration, and the type of land use are defined as physical "inventory data" that directly describe the intervention in nature. The described change is then "characterized" in its environmental impact (Lindner et al. (in preparation), p. 20).

The method takes into account that the impacts on biodiversity (both positive and negative) must be determined locally, and it also considers that not all processes can be quantified equally well in terms of their impacts on biodiversity. Therefore, the method precedes the actual characterization with a significance analysis, in which the relevance of the processes regarding their impacts on biodiversity is classified into three categories (high, medium, low relevance):

- *Processes with high relevance are characterized using the detailed potential field method, in which the biodiversity potential of individual processes is estimated based on ecoregion-specific input parameters*
- *Processes with medium relevance are roughly estimated using the BDP method by De Baan et al. (2013)*
- *Processes with low relevance are neglected*

*The indicator "**Quality of an area in relation to biological diversity**" is described as follows:*

$$Q_{BV,i} = EFi * BP_i$$

$Q_{BV,i}$: Value of biodiversity on a and area element in region i

EF_i : Ecoregion factor of region i

BP_i : Height of biodiversity potential in region i

EF_i is a predetermined factor for each ecoregion i. BP_i is a function with approximately 5 to 10 input parameters x_{ij} , which are weighted and aggregated. The value of BP_i ranges between 0 and 1:

$$BP_i = (x_{ij}) \in (0; 1)$$

x_{ij} : Parameter j to describe the biodiversity in region i

Lindner et al. (in preparation), S. 24

The quality of the area is determined for each relevant land-use process: for processes with high relevance, the detailed potential field method is used, and for processes with medium relevance, the method of De Baan et al. (2013) is applied. The results of the rough estimation are aggregated with those of the detailed estimation.

The biodiversity impacts obtained for each input parameter are aggregated with equal weighting to form the biodiversity potential. The biodiversity potentials of different ecoregions are multiplied by an ecoregion factor and then added to account for the varying value of different ecoregions. The ecoregion factor is derived from the WWF Wildfinder Database (WWF 2006). The resulting value indicates the potential by which biodiversity is negatively affected by the product/service. The lower the biodiversity potential, the more negative the impact on biodiversity. For the Handprint logic, the biodiversity potential must be multiplied by -1 and subtracted from 1.

Results of the case study biodiversity

In the biodiversity case study, Irish milk production is compared with milk production in the Allgäu region. The basis for this comparison includes data from Irish dairy farms and data from two dairy farms in the Allgäu region (Eberle & Lindner 2015), which are about twice the size of an average Irish farm, with approximately 140 cows.

It was assumed that the ecoregion in which the Irish farms are located (Celtic Broadleaf Forests, PA0409) behaves similarly in terms of measuring potential impacts on biodiversity to the ecoregion 'Western European Broadleaf Forests' (PA0445), where the comparison farms in the Allgäu are

situated. This means that the same input parameters were used for calculating the biodiversity potential. These parameters include pesticide use, nitrogen surplus, biomass use, the proportion of small structural elements, and the number of varieties/breeds used (Lindner et al., in preparation).

The calculation of biodiversity potential is done using the potential field curves of each input parameter (see Appendix to the Food Case Studies) for the defined functional unit of 100 kg FPCM⁴. The obtained biodiversity potential is then multiplied by -1 according to the Handprint logic, representing positive impacts with positive numbers and negative impacts with negative numbers.

As a result, biodiversity impacts for Irish milk are -70.41 per functional unit, and an average of -173.45 for milk from the Allgäu, with impacts on the biodiversity of the considered farms in the Allgäu ranging from -126.7 to -220.2. This means that the impacts on biodiversity in Irish milk production at Kerrygold's supplier farms are significantly lower than at the considered farms in the Allgäu.

Biodiversity impacts are mainly influenced by the type of feeding, especially a high proportion of green fodder feeding causes lower biodiversity impacts. The better nitrogen balance and higher proportion of small structural elements in Ireland also contribute to lower biodiversity impacts.

In terms of the question posed in the case study (sharpness of the result), it can be stated that the applied method is suitable for generating results that reveal a real difference in the systems under consideration. However, it should be noted that biodiversity impacts are generally attributed to the Footprint.

Regarding the question of whether potentially positive sustainability impacts can be determined, it must be noted that – unlike indicators related to company activities (e.g., employment conditions, management activities, reporting) – an indicator directly connected to the material flows of the assessed product is challenging to evaluate with the proposed approach (Fuzzy Set Method, Potential Field Theory). The reason for this is that it would need to be determined what proportion the respective product, for instance, contributes to greenhouse gas emissions or biodiversity impacts. This means that a different method of evaluation must be found for indicators directly related to material flows. In the project, this question remains unanswered, as the indicators mostly represent potentially negative sustainability impacts. Nevertheless, a comprehensive sustainability assessment always requires considering potential positive and negative sustainability impacts to obtain a complete picture.

⁴ FPCM: fat and protein corrected milk (IDF 2010)

2.4.1.2 Case study reference value

In the case study on the reference framework, an examination was conducted to determine the differences in the assessment results of selected social indicators due to different reference frameworks.

In a life cycle assessment according to ISO 14040/44, the reference framework typically consists of the material flows required to produce the defined functional unit. Additionally, when considering potential social impacts associated with the product system, the required labour hours to produce the functional unit are commonly used as the reference framework.

Unlike the consideration of ecological impacts directly associated with material flows, social impacts are often only indirectly connected to the number of labour hours. An example could be management activities, which are usually conducted for the entire company and, therefore, are related to both the material flows of the functional unit and the labour hours of the functional unit.

In the case study, an investigation was conducted into the variations in results when different reference frameworks are applied. Reference frameworks based on cost contributions of the companies involved along the life cycle, various value-added contributions, or the level of influence were not examined here, as these were already compared by Dreyer (2009) using labour hours as the reference framework.

The influence of the following reference frameworks on the assessed results in social indicators was examined:

- Material flows
- Labour hours
- Life cycle phases: Each actor group was assigned to a life cycle phase - farmers to the "agricultural production" phase, the dairy to the "processing" phase, and Ornua Deutschland GmbH to the "packaging/distribution" phase. Each of these phases was equally weighted, meaning each phase was mathematically assigned the value "one."
- Number of actors: The number of involved actors was used as the basis for weighting. This includes 14,000 actors at the agriculture level, 3 actors at the dairy level, and 1 actor at the Ornua Deutschland level.

Furthermore, it is important to note that the production of butter generates by-products at various stages. Therefore, various allocation methods were applied, and their impact in combination with the choice of the reference framework was examined. At the agricultural production level, economic allocation was chosen for all cases. This results in allocating 85% of the impacts to milk, with the remaining attributed to calves and meat. At the milk processing level, three different allocation methods were applied for the allocation to butter, buttermilk from butter production, and skim milk from cream centrifugation:

- 100% allocation to butter production (100%)
- Allocation based on milk dry matter (MDM): This results in allocating 32% of the impacts to butter (Feitz et al., 2005).
- Economic allocation based on market prices for butter, buttermilk, and skim milk results in allocating 33% of the impacts to butter (economic) (Nilsson et al., 2010).

For the case study, the two indicators "Coverage of sustainability (risk) management" and "Coverage of social security systems" were used. The manifestation of the indicator was examined at the actor level, i.e., the involved company (agricultural enterprise, dairy, packager/distributor). Not all actors involved along the life cycle were included in the case study; for example, transport actors were not recorded nor were packaging manufacturers. The evaluation of the two indicators at the level of individual actors yields the results presented in Table 3 below. These results are not yet related to the reference framework.

Additionally, three hypothetical indicators were examined to analyse the influence of the reference framework and allocation methods on extreme manifestations of assessment results.

Table 3: Assessment results of the indicators at each stage

	Ornua Germany	Dairy Ireland	Farms Ireland
Sustainability (risk) management	0,42	0,33	0,42
Social security systems	0,67	0,56	0,56
Hypothetical indicator 1	0	0	1
Hypothetical indicator 2	1	0	0
Hypothetical indicator 3	0	1	0

When these assessment results are then related to the various examined reference values, the image depicted in Figure 5 emerges for the four selected reference values.

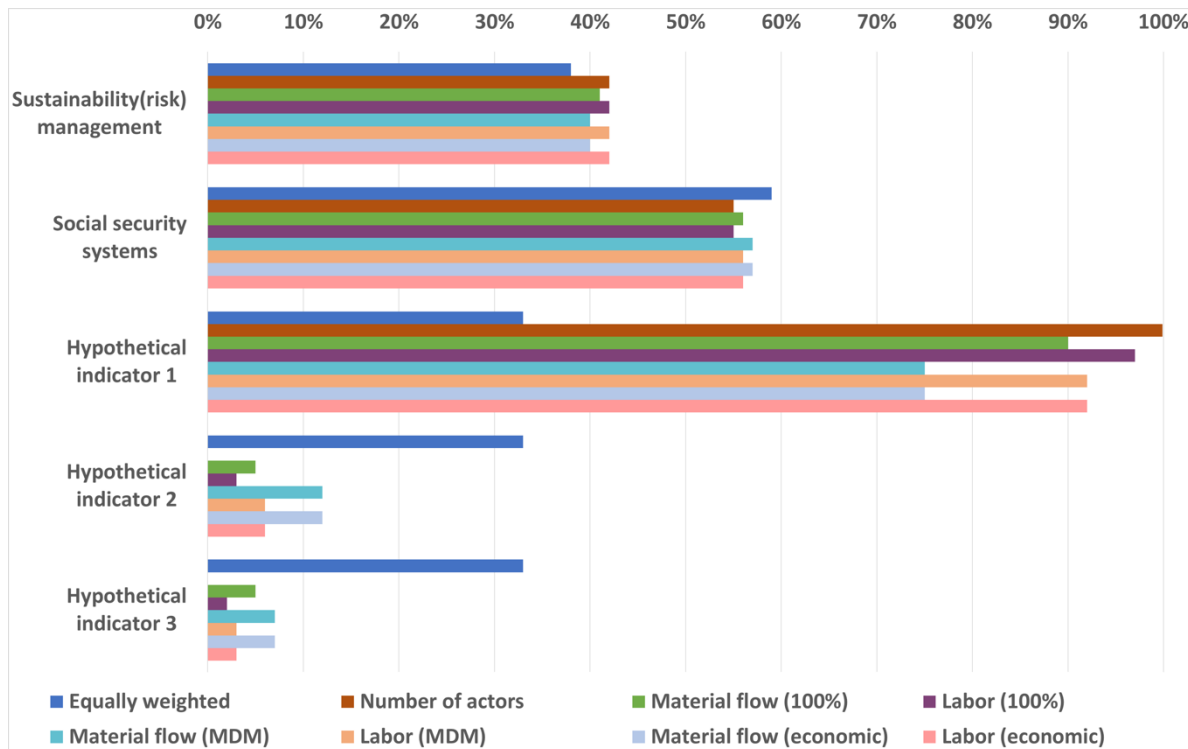


Figure 5: Impact of different reference values and allocation methods on the assessment results

The results indicate that the equal weighting of individual life cycle phases and the number of actors have a significant impact on the overall assessment outcome. This is particularly evident in the sensitivities calculated using three hypothetical indicators. The difference in results between the life cycle phase as a reference value and the number of actors can be up to 100%. Depending on the allocation method, the difference between material flows and working hours ranges between eight and 80 percent. Additionally, it is observed that using equally weighted life cycle phases as a reference value levels out significant differences in assessment results between the life cycle phases (comparison of hypothetical indicators 1-3) and is therefore not a suitable reference value. Similar observations can be made for the reference value “number of involved actors”: here, the mere number of actors significantly influences the results (comparison of hypothetical indicators 1-3). However, these drastic differences are not as visible in the two real indicators due to the minor differences in assessments between life cycle phases or at the actor level. Nevertheless, the difference between the life cycle phase and the number of actors (minimum and maximum) is seven percent, while between material flows and working hours, it is only between one and two percent.

Although differences exist between using working hours and material flows as reference values⁴, it is challenging to decide which one is the better reference value based on the results. This is primarily due to the relatively independent nature of the indicators considered from both reference values.

2.4.1.3 Case study weighting

Within this case study, the influence of weighting individual topics of an indicator on the overall assessment of the indicator was examined. Two indicators, "Coverage of sustainability (risk) management" and "Coverage of social security," were chosen for this purpose. In the first case, there was an equal weighting in the evaluation among the various thematic areas of the respective indicator. In the second case, certain thematic areas were considered "essential" and therefore were rated as twice as important as the remaining thematic areas.

The "Sustainability (risk) management" indicator comprises thirteen subject areas: "Agriculture" (SDG 2.4), "Water" (SDGs 6.5), "Freshwater ecosystems" (SDG 6.6), "Use of natural resources" (SDG 12.2), "Handling chemicals" (SDG 12.4), "Waste management" (SDG 12.5), "Climate change" (SDG 13.2), "Marine ecosystems" (SDG 14.2), "Terrestrial biodiversity" (SDGs 15.1-15.5), "Patents on genetic resources" (SDG 15.6), "Poverty reduction" (SDG 15.9), "Corruption prevention" (SDG 16.5), and the "Promotion of environmental technologies in developing countries" (SDG 17.7). Since "Poverty reduction" was not queried in the case study, the evaluation is limited to 12 thematic areas.

Table 4: Indicator "Coverage of sustainability (risk) management" – equal weighting of topics

Subject	Ornua Germany	Dairy Ireland	Farms Ireland
Agriculture (soil)	0,00	0,00	0,08
Water	0,08	0,08	0,08
Freshwater ecosystems	0,00	0,08	0,08
Natural resources	0,08	0,00	0,00
Chemicals	0,08	0,08	0,00
Waste	0,08	0,08	0,00
Climate change	0,08	0,00	0,00
Marine ecosystems	0,00	0,00	0,08
Terrestrial biodiversity	0,00	0,00	0,08
Patents on natural genetic resources	0,00	0,00	0,00
Poverty reduction	X	X	X

Corruption prevention	0,08	0,00	0,00
Promotion of environmental technologies in developing countries	0,00	0,00	0,00
TOTAL	0,50	0,33	0,42

Table 5: Indicator “Coverage of sustainability (risk) management” – double weighting of the highlighted yellow thematic areas

Subject	Ornua Germany	Dairy Ireland	Farms Ireland
Agriculture (soil)	0,00	0,00	0,06
Water	0,12	0,12	0,12
Freshwater ecosystems	0,00	0,12	0,12
Natural resources	0,06	0,00	0,00
Chemicals	0,06	0,06	0,00
Waste	0,06	0,06	0,00
Climate change	0,12	0,00	0,00
Marine ecosystems	0,00	0,00	0,12
Terrestrial biodiversity	0,00	0,00	0,12
Patents on natural genetic resources	0,00	0,00	0,00
Poverty reduction	X	X	X
Corruption prevention	0,06	0,00	0,00
Promotion of environmental technologies in developing countries	0,00	0,00	0,00
TOTAL	0,47	0,35	0,53

The indicator of social security consists of nine subject areas, eight of which are predefined, and an additional subject area can be freely added. These include health insurance, unemployment insurance, pension, nursing care insurance, severance pay, accident insurance, disability insurance, and survivor benefits.

Table 6: Indicator "Coverage of social security" – equal weighting of subject areas

Topic of social security	Ornua Germany	Dairy Ireland	Farms Ireland
Health insurance	0,11	0,00	0,00
Unemployment insurance	0,11	0,11	0,11
Retirement/ Pension	0,11	0,11	0,11
Long-term care insurance	0,11	0,00	0,00
Severance pay in the event of termination of the employment relationship	0,00	0,00	0,00
(Group) accident insurance	0,11	0,11	0,11
Occupational disability	0,00	0,11	0,11
Survivors' benefits	0,00	0,11	0,11
Other: Pension insurance from 5 years of service, inheritable	0,11	0,00	0,00
TOTAL	0,67	0,56	0,56

Table 7: Indicator "Coverage of social security" - double weighting of the highlighted subject areas

Topic of social security	Ornua Germany	Dairy Ireland	Farms Ireland
Health insurance	0,17	0,00	0,00
Unemployment insurance	0,17	0,17	0,17
Pension	0,17	0,17	0,17
Long-term care insurance	0,08	0,00	0,00

Severance pay in the event of termination of the employment relationship	0,00	0,00	0,00
(Group) accident insurance	0,08	0,08	0,08
Occupational disability	0,00	0,08	0,08
Survivors' benefits	0,00	0,08	0,08
Other: Pension insurance from 5 years of service, inheritable	0,08	0,00	0,00
TOTAL	0,75	0,58	0,58

In both cases, it becomes evident that the weighting has an impact on the results: for the "Social security coverage" indicator, the chosen change in weighting alters the magnitude of the evaluation. All three actors receive an overall more positive assessment, and the gap between Ornuu and the farms or dairies increases. For the "Sustainability (risk) management coverage" indicator, the change in weighting also alters the ranking of the assessed actors. This means, as with other weightings, that the chosen weighting must be well-justified and transparently presented.

Conclusions from the food case studies:

The initial evaluation indicates that the data return takes a long time - as is usual with Life Cycle Assessments - and always longer than expected and agreed upon. Additionally, data queries are never fully completed, leaving data gaps for which a procedure must be defined on how to handle them. This means that the method always faces the challenge of delivering precise results based on the highest possible specific primary data share while remaining practical. The results of the food case studies indicate that the initially proposed indicators in the project may require some adjustments. It is also necessary to discuss the assessments (potential field curves) for each indicator and involve additional stakeholders in this process.

There are still open methodological questions, such as the connection between the Sustainable Development Goals (Nexus) and dealing with potential double counting in this context. Questions about possible aggregation into a sustainability assessment number and handling conflicts of interest in this context need to be addressed to complete a comprehensive evaluation (Handprint and Footprint).

Appendix to case studies food: potential field curves of input parameters for the ecoregion 'Western European Broadleaf Forests'

The following are the potential field curves of the individual input parameters (Lindner et al., in preparation):

- Biomass utilization: The input parameter biomass utilization describes the proportion of existing biomass removed from the area. This is particularly relevant for biodiversity because net primary production forms the basis for all subsequent trophic levels. This can be expressed by the indicator proportion of net primary production extraction.

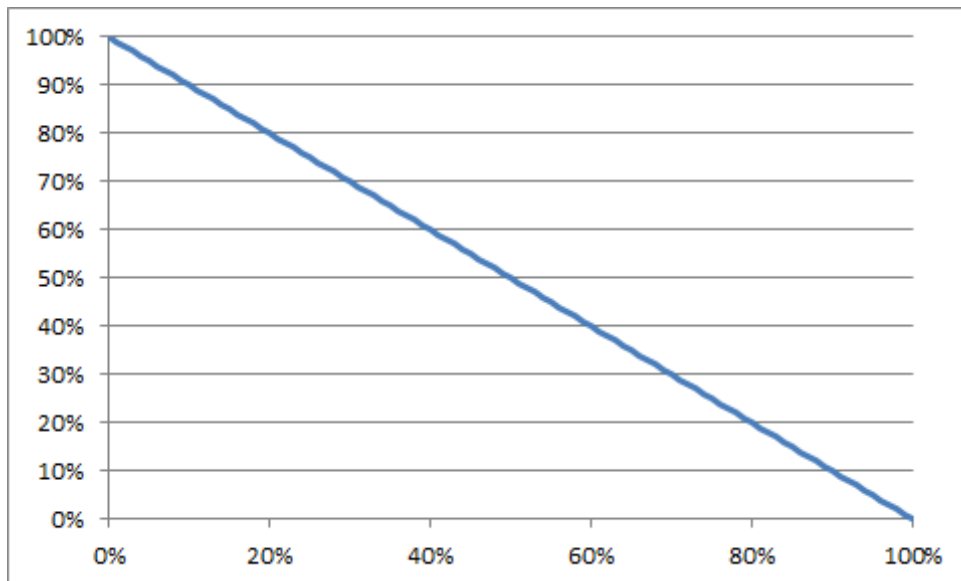


Figure 6: Biomass utilization; x-axis: biodiversity impact, y-axis: % biomass extraction/ha*year

- Nitrogen balance: The use of fertilizers in agriculture aims to achieve better yields. For this purpose, nutrients-usually nitrogen, phosphorus, and potassium-are supplied to crops in absorbable form.

Relevant to biodiversity is primarily the excessive application of fertilizers, leading to areas being over-fertilized (eutrophication). In cases of over-fertilization, the focus is typically on the inputs of nitrogen and phosphorus. Eutrophication can occur both in aquatic ecosystems (aquatic eutrophication) and in terrestrial ecosystems (terrestrial eutrophication). Anthropogenically induced nutrient inputs into terrestrial ecosystems can result in harmful effects. For example, nitrogen-loving plants may better utilize the abundant nitrogen supply than other species, growing faster and more expansively, outcompeting other species. Consequently, the species composition of the ecosystem changes, often reducing the number of species while increasing the number of individuals of certain species.

This can be expressed by the Nitrogen balance indicator.

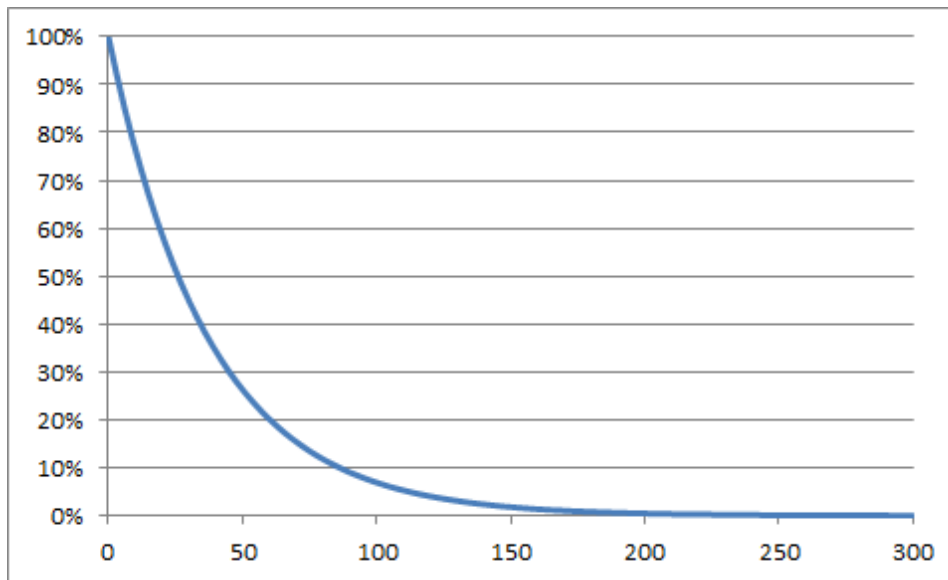


Figure 7: Nitrogen balance parameters; x-axis: biodiversity impact, y-axis: N-Excess/ha*a

- Pesticide use: The application of pesticides always has negative effects on biodiversity, as these substances function, depending on their spectrum of action, to eliminate unwanted plants (herbicides), animals (insecticides, acaricides, rodenticides, etc.), or fungi (fungicides). Despite adhering to rules of good agricultural practices, active ingredients can be transported into adjacent ecosystems, leading to alterations in species composition and potentially the number of species present. Ecotoxicity potential is employed as an indicator for pesticide use. Various models have been proposed for determining the toxicological impacts of substances on the environment. In 2008, the UNEP/SETAC developed a scientific consensus model for this purpose, which is also employed here: USEtox (Rosenbaum et al. 2008, Fantke et al. 2015).

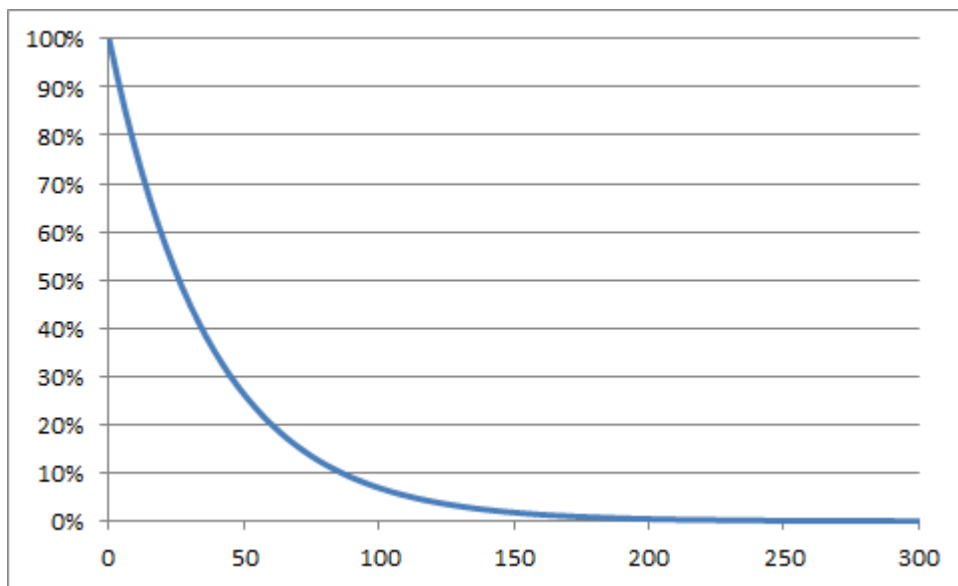


Figure 8: Pesticide use parameters; x-axis: biodiversity impact, y-axis: CTUe/ha*a

- Structural Diversity 1: The input parameter "structural diversity 1" determines the proportion of small structures on the area. Small structures in agriculturally utilized areas have a positive impact on biodiversity. In Germany, a "Directory of Regional Small Structure Shares"⁵ is maintained by the Julius Kühn Institute (JKI), from which both the current status and target values at the municipal level can be derived. The indicator used is the so-called habitat index, representing the proportion of small structures on the agricultural area excluding permanent grassland⁶.

⁵ <http://www.jki.bund.de/de/startseite/fachinformationen/pflanzenschutz/pflanzenschutzverfahren/kleinstrukturen.html>; Status: 21. Januar 2015

⁶ http://www.jki.bund.de/fileadmin/dam_uploads/_SF/kleinstrukturen/Beschreibung%20der%20Methode%20zur%20Ermittlung%20der%20Kleinstrukturen.pdf; Status: 21. Januar 2015

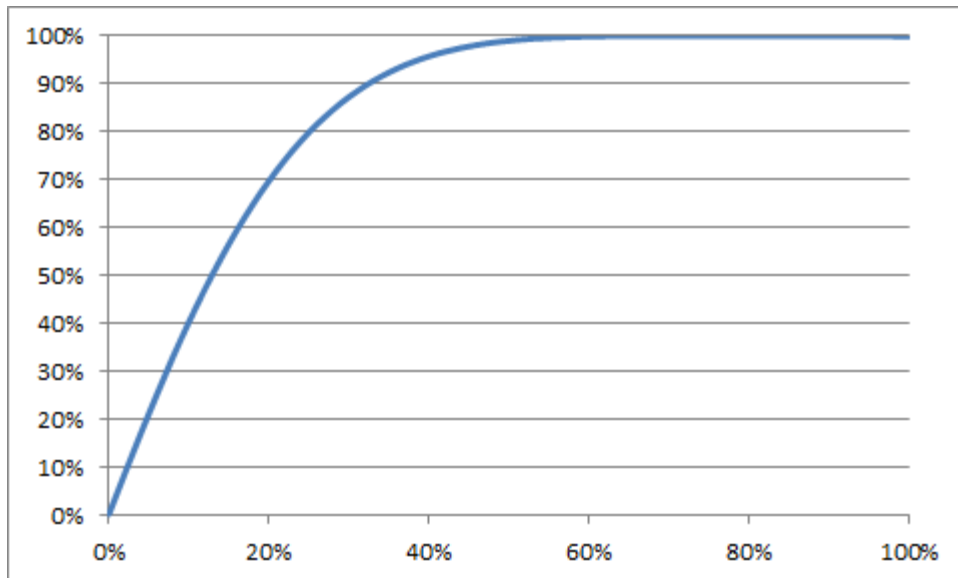


Figure 9: Structural diversity parameters 1; x-axis: biodiversity impact, y-axis: % small structural elements/ha*yr

- Structural Diversity 2: The input parameter structural diversity 2 describes the number of agriculturally used species (agrobiodiversity), i.e., the species that are sown. Permanent grassland is not considered in this context.

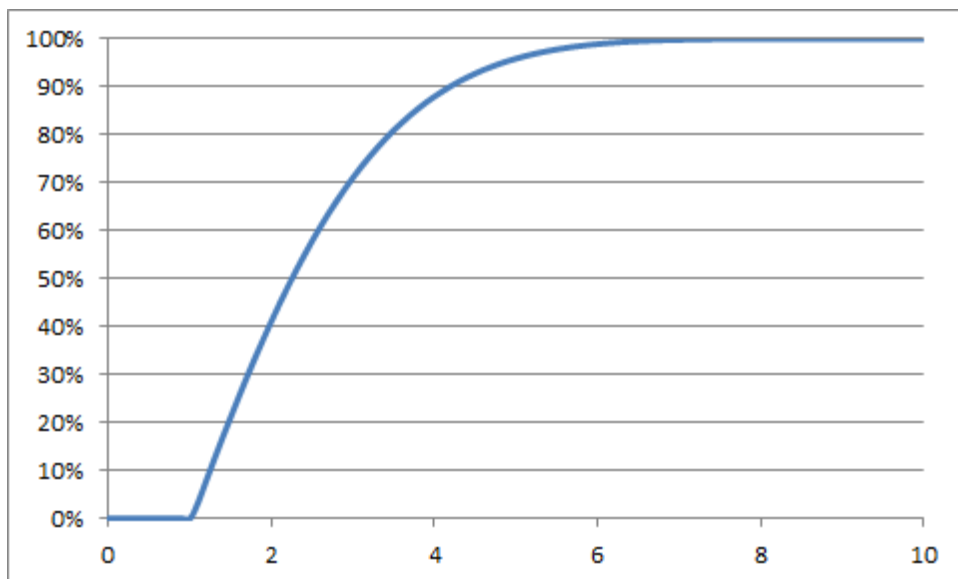


Figure 10: Structural diversity parameters 2; x-axis: biodiversity impact, y-axis: number of varieties on the farmyard

2.4.2 Consumer goods case studies

Authors: Janpeter Beckmann, Christoph Herrmann

Selection of the Case Study Partner

For the consumer goods industry, the decision was made to focus on the product "Faire Maus" – a computer mouse – from Nager IT e.V.. The connection with the case study partner was established through a FONA networking event and the BiNa project, in which Nager IT is involved as an industrial partner. The computer mouse is a complex product with an internationally interconnected value chain, composed of a variety of materials and components. This provides the opportunity to apply the Handprint methodology comprehensively and explore a wide range of scenarios to examine methodological aspects. The nature of the product or production suggested interesting Handprints a priori (e.g., casing made of bioplastic, manufacturing in an integration workshop), as well as the company's commitment to a transparent and fair supply chain. Furthermore, Nager IT e.V. demonstrated high motivation to participate in the case study.

The Product "Faire Maus"

Computer mice are electronic products composed of various components produced and distributed in globally connected value chains. The production of these components requires raw materials such as petroleum and ores from different parts of the world⁷. The global value chains of electronic products are often opaque and difficult to trace due to a lack of information availability. Additionally, raw materials are often extracted in countries with lower environmental and social standards or where these standards are not adequately enforced. The same applies to the production of components.

The differences of the "Faire Maus" compared to a conventional computer mouse lie not only in the continuously advancing transparency in the supply chain and the endeavour to make it as "fair" as possible but also in the substitution of certain materials to enhance the sustainability of the product from ecological and social perspectives. The casing, including the buttons of the "Faire Maus," is made from bioplastic. Additionally, the scroll wheel is crafted from local and certified wood. The solder used for assembling the circuit boards is made from recycled material and has been confirmed as "fair" by an initiative dedicated to fair soldering ("Fairlötet"), which was initiated by Nager IT.

⁷ A variety of raw materials are required to manufacture a computer mouse, including crude oil, silicon, copper, tin, nickel, aluminum, titanium, iron and silver, copper, tin, nickel, aluminum, titanium, iron and silver.

The final assembly takes place in an integrative workshop in Regensburg, where individuals with mental health conditions work, receiving individually tailored work requirements and support. Additionally, other components come from Germany, manufactured according to local social and environmental standards. This applies to resistors, capacitors, electrolytic capacitors, circuit boards, screws, packaging (from Austria), labels, dyes, and solder.

The "Faire Maus" is priced at approximately thirty euros.

Comparable conventional computer mice are available for purchase for under ten euros. The "Faire Maus" is produced by Nager IT e.V. as part of an ongoing pioneering project. The motivation for the fair computer mouse project stems from the opaque electronics industry, where supply chains and corresponding working conditions are challenging to ascertain. Simultaneously, a multitude of sustainability risks are known (see Table 8)⁸.



Figure 11: Model "Faire Maus" by Nager IT

The fair computer mouse aims to serve as a pioneering effort, inspiring other companies to reconsider their practices, similar to what has already occurred in other global product categories such as tea, coffee, or cocoa. Major companies have already incorporated fair products into their portfolios in these areas. Nager IT e.V.'s vision is to lead the way in global IT production with fair working conditions, setting an example for all stakeholders.

⁸ Avilez, 2003; FIF, 2018; Hütz-Adams, 2012; Odrich, 2014; Smirnova, 2015

Table 8: Examples of social and environmental sustainability risks in the extraction of raw materials and manufacturing of conventional computer mice

	Sustainability risks	
	Social	Environment
Raw materials phase (the production of a computer mouse requires a variety of raw materials, including crude oil, silicon, copper, tin, nickel, aluminum, titanium, iron and silver)	• Social grievances in the mines (e.g. in Africa); including violent conflicts, dangerous working conditions with many accidents • Mine workers in Africa are often unable to feed their families on the wages they earn • Harmful working conditions due to toxic hazards (e.g. arsenic dust during copper mining in Chile) • Illegal small-scale mining in particular has negative consequences for people and nature due to the lack of standards	• Generally massive environmental pollution for cost reasons; including leaking pollutants such as methane, lead, arsenic and impurities (e.g. nickel sludge) • Illegal small-scale mining with a lack of environmental standards • Bauxite deposits (for aluminum) are largely located in the tropical belt; mining is associated with the destruction of rainforest • Damage to ecosystems (e.g. red mud from aluminum smelting in Brazil, release of sulfur dioxide from nickel processing in Russia)
Manufacturing	• Poor working conditions and non-compliance with ILO core labor standards in Asian factories, including non-compliance with health regulations, overtime, lack of employment contracts and inhumane accommodation	• Significant amounts of electricity and energy lead to high greenhouse gas emissions

As an electronic product, the computer mouse is composed of various components and parts, which can be categorized into internal and external components. Starting from the mass of the mouse, the USB cable, circuit board, buttons, and the casing consisting of upper and lower shells are of particular importance. The following tables provide an overview of the components, their respective production locations, and masses in grams.

Table 9: Components of the internal structure of the “Faire Maus”. Schnurr (2017)

Component	Production location	Mass [g]
Chip	Philippines	0,516
Wire bridges, total	UK	0,220
Rotary potentiometer Z1, 50 mΩ	China	0,505
Elkos capacitor C2, 10 μF	Germany	0,661
Elkos capacitor C5, 4,7 μF	Germany	0,473
Capacitor C3, 0,1 μF	Germany	0,198
LED	Japan	0,275
Circuit board	Germany	7,670
Lens	Malaysia	1,200
2 switches, each	China	1,000
SMD Capacitors C9/10, 22 pF, each	Israel	0,004
SMD resistors R BIN, 75 Ω R7, 30 kΩ R8/9, 22 Ω R10, 15 kΩ R11, 6,8 kΩ, each	Germany	0,004
USB cable 1.5 m	China	19,635

Table 10: Components of the external life of “Faire Maus”. Schnurr (2017)

Component	Production location	Mass [g]
Label	Germany	n.s.
2 stands, each	China	0,017
Keys, total	Germany	8,500
Top shell Housing	Germany	21,500
2 floor screws, each	Germany	0,440
2 screw covers, each	Germany	0,540
Scrollwheel	Germany	2,300
Base shell Housing	Germany	19,500

Overview of the Entire Supply Chain is provided in Figure 12 in the following section “Execution”.

Execution

The supply chain of the fair mouse is highly complex due to the multitude of components. Therefore, supply chain partners were initially divided based on their relevance, with the aim of covering as many components of the computer mouse as possible and identifying key companies for

each component chain. A primary data collection was carried out for these companies, while additional companies were evaluated as much as possible through desktop research.

Key companies, such as Retex, assemblers of individual components, were empirically analysed through primary data collection. A comprehensive data collection took place with a visit to Retex. A personal meeting with representatives from Stannol and Fairlötet (solder wire) occurred, and the desired data could be almost completely obtained. An intensive telephone exchange with IfBB Hochschule Hannover (bioplastic) facilitated a complete evaluation of Beoplast (casing) through primary data collection. Frolyt (electrolytic capacitors) also filled out the questionnaire.

Despite intensive efforts, the following companies could not be obtained for primary data collection, but were evaluated through desktop research and, in some cases, additional short interviews:

- Microtech electronic (resistors)
- Hofmann LP (circuit board)
- WIMA (capacitors)
- EJOT (screws)
- Hauck (wooden wheel)

Additionally, a risk analysis of the procurement and processing of mineral raw materials was conducted through desktop research concerning ecological and social factors, as well as potential Handprints.

The following diagram of the supply chain illustrates the extent and nature of the data collection. Companies with green markings underwent primary data collection, while companies and raw materials with orange markings were covered through desktop research (partially with supplementary short interviews). White fields represent companies that were not investigated (no available data, no responses to inquiries, etc.).

2.4.2.1 Case study substitution of materials

Solder consists of over 95 percent tin, with additional components including copper, nickel, cobalt, and germanium. A flux is also required, comprising rosin (e.g., tree resin from Brazil), petroleum-based wax, and other chemicals. Since the raw materials are often not purchased individually but are already contained in specific alloys, direct contact with the source of raw materials is often not available. Therefore, achieving a "fair" design of the supply chain is challenging. Nager IT, in collaboration with the Fairlötet initiative, has taken on this challenge. In fair solder, tin, cobalt, nickel, and part of the copper (more than 95 percent in total) are made from recycled materials. Waste that would otherwise have been processed into lower-quality products ("downcycling") was recycled. Avoiding downcycling can also assume an extended presence of the metal in the lifecycle of one or more products.

Regarding social factors, the procurement and use of conventional tin entail the following risks, which are intended to be avoided using fair tin⁹:

- Working conditions are generally harmful to health due to contact with harmful dust and lack of occupational safety. Disability and fatalities also occur due to accidents, for example, in poorly secured mines or open-pit mining below the water table
- Payment is not subsistence and does not allow workers to lift themselves out of poverty. Child labour is also common
- Tin mining is not carried out in coordination and balance among various interest groups but instead involves the expulsion and repression of the population. This frequently leads to environmental destruction and the loss of livelihoods for people relying on agriculture and fishing
- The absence of basic worker's rights such as freedom of association

For the "Faire Maus," an alloy is used with tin and germanium sourced from a supplier who has provided a certificate indicating that the tin is conflict-free, meaning it is not sourced from conflict or high-risk areas. The rosin (flux resin) is obtained in Brazil, and the supplier has given written assurance that this is done in compliance with social and environmental minimum requirements. According to Nager IT, a more sustainable rosin is currently not available.

Half of the tin comes from a recycler in Germany, primarily processing electronic waste. The other half, along with cobalt, nickel, and part of the copper, comes from a contaminated solder bath. The combination of these two sources has produced a material that complies with industry legal standards and quality requirements¹⁰.

A detailed analysis of environmental indicators has shown that the substitution process initially comes with reduction potentials for certain negative environmental impacts, thereby positively affecting some environmental indicators, such as land use, acidification potential, marine eutrophication potential, and particulate matter potential¹¹.

The substitution of tin with fair tin or, in part, recycled tin reveals – in addition to the mentioned potential positive environmental effects – the following initial (positive) sustainability impacts in the areas of economy, governance, and social aspects:

⁹ <https://fairloetet.de/wp-content/uploads/2018/10/Whitepaper.pdf>

¹⁰ <https://www.nager-it.de/static/pdf/Fairzinn.pdf>

¹¹ The results of the detailed analysis of the reduction potential can be found in Schnurr 2017. transferability of methodological components of life cycle assessment to the Handprint of products; can be viewed at the Chair of Energy Systems and Energy Economics at Ruhr University Bochum (<http://www.lee.ruhruni-bochum.de/>)

- Contribution to waste reduction and increased resource efficiency by using recycled material.
- Positive social effects, particularly in the realm of occupational safety, stemming from the use of certified materials and collaboration with the "Fairlötet" initiative.
- Governance effects such as generated transparency and active communication of sustainability aspects due to applied standards and the public engagement created through the "Fairlötet" cooperation.

In addition, an examination will be conducted to assess the extent to which the methodology allows the capture and evaluation of potential additional (relevant) effects. These are derived from Nager IT's efforts to establish a

- transparent and fair supply chain, as well as
- dignified working conditions. For this purpose, Nager IT aims to meet at least the following points from the International Labour Organization's (ILO) demands:
 - Regulated working hours
 - Adequate compensation
 - Health protection
 - Social security
 - Freedom of assembly
 - Elimination of exploitative child labour and forced labour
 - Responsible supply relationships (fair prices, "timely communication of wishes," long-term business relationships)

From the indicator pool of the Handprint methodology, the following indicators are initially considered for the assessment and evaluation of effects due to their thematic proximity or direct relevance to the (sub-) goals of the SDGs.

Table 11: Selection of indicators relevant to the context of the case study

Indicator	SDGs	
Contribution to the reduction of waste (Economy)	12.4	By 2020, achieve the environmentally sound management of chemicals and all wastes throughout their life cycle, in accordance with agreed international frameworks, and significantly reduce their release to air, water and soil in order to minimize their adverse impacts on human health and the environment
	12.5	By 2030, substantially reduce waste generation through prevention, reduction, recycling and reuse

Indicator	SDGs	
Contribution to resource efficiency (efficient use of resources, reuse/recycling content, etc.) (economy)	8.4	Improve progressively, through 2030, global resource efficiency in consumption and production and endeavour to decouple economic growth from environmental degradation, in accordance with the 10-year framework of programmes on sustainable consumption and production, with developed countries taking the lead.
	12.2	By 2030, achieve the sustainable management and efficient use of natural resources.
	12.3	By 2030, halve per capita global food waste at the retail and consumer levels and reduce food losses along production and supply chains, including post-harvest losses.
	12.5	By 2030, substantially reduce waste generation through prevention, reduction, recycling and reuse.
Commitment to the development of sustainability standards (governance)	16.6	Develop effective, accountable and transparent institutions at all levels
Compliance with sustainability standards	12.8	By 2030, ensure that people everywhere have the relevant information and awareness for sustainable development and lifestyles in harmony with nature
	12.6	Encourage companies, especially large and transnational companies, to adopt sustainable practices and to integrate sustainability information into their reporting cycle.
	16.6	Develop effective, accountable and transparent institutions at all levels
Integration of sustainability information into reporting	12.8	By 2030, ensure that people everywhere have the relevant information and awareness for sustainable development and lifestyles in harmony with nature
	12.6	Encourage companies, especially large and transnational companies, to adopt sustainable practices and to integrate sustainability information into their reporting cycle.
Active communication of sustainability aspects to users	12.8	By 2030, ensure that people everywhere have the relevant information and awareness for sustainable development and lifestyles in harmony with nature
Average wage of workers compared to minimum wage and living wage	1.1	By 2030, eradicate extreme poverty for all people everywhere, currently measured as people living on less than \$1.25 a day.

Indicator	SDGs	
	8.5	By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.
Expenditure for social security	1.3	Implement nationally appropriate social protection systems and measures for all, including floors, and by 2030 achieve substantial coverage of the poor and the vulnerable.
	8.5	By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.
Ratio of men's to women's pay (at different salary levels)	8.5	By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.
Number of cases of child labour along the product life cycle Number of cases of forced labour along the product life cycle	8.7	Take immediate and effective measures to eradicate forced labour, end modern slavery and human trafficking and secure the prohibition and elimination of the worst forms of child labour, including recruitment and use of child soldiers, and by 2025 end child labour in all its forms.
Promoting diversity and equal treatment	8.5	By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.
Sustainability (risk) management in the company and in the supply chain	12.2	By 2030, achieve sustainable management and efficient use of natural resources.
	12.6	Encourage companies, especially large and transnational companies, to adopt sustainable practices and to integrate sustainability information into their reporting cycle.
	16.3	Promote the rule of law at the national and international levels and ensure equal access to justice for all.

Not all the listed indicators could be developed during the project; some are still in the discussion phase. Therefore, the indicators “*Contribution to resource efficiency*” and “*Expenditures for social security and sustainability (risk) management in the company and in the supply chain*” cannot be considered in the analysis.

Analysis of the indicators and potential goal conflicts

a) Indicator: Contribution to waste reduction

The impact of using recycled solder on this indicator is very low, as the mass share of solder in the entire computer mouse can be almost negligible. This indicator solely considers the material

composition. From the perspective of materiality, meaning the relevance to the impacts, this indicator would likely not be considered in practical application of the method.

Furthermore, from the perspective of materiality, the question arises as to the extent to **which other sustainability effects** related to the sub-supply chain of solder, as discussed below, should be classified as relevant. This question will be discussed in the summary of results.

Contribution to reducing waste/waste volumes

SDG-REFERENCES: 12.4., 12.5.

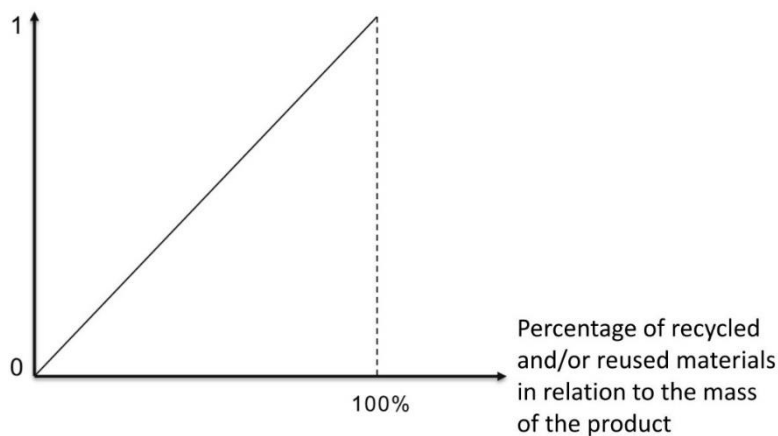


Figure 13: Indicator “Contribution to waste reduction”

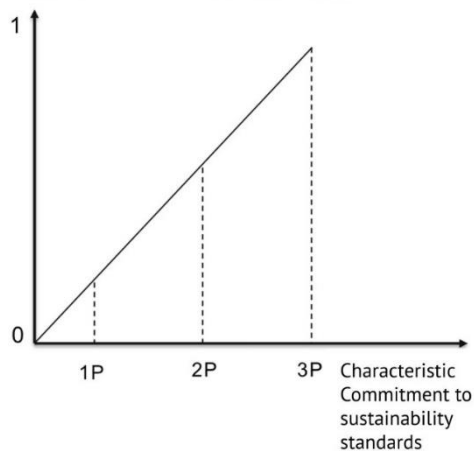
b) Indicator: Engagement for the development of sustainability standards

The endeavour of Nager IT to produce a “fair mouse” and to use fair solder as part of this endeavour can be captured through the indicator “*Engagement for the development of sustainability standards*”: Nager IT actively participates in the development of the “Fairlötet” initiative, which works on sustainability standards in the field of solder. Since the initiative’s results are not limited to Nager IT, the initiative is publicly accessible, and third parties are involved, Nager IT would receive full points for this indicator.

It is worth noting that Nager IT could earn additional points through participation in other initiatives, but these would not carry much weight due to the achieved maximum score. It would be worth exploring whether the methodology aims to reward comprehensive engagement by companies, i.e., participation in multiple initiatives.

Commitment to the development of sustainability standards

SDG-REFERENCES: 16.6.



Points system proposal

- 0 points – No commitment to the development of sustainability standards
- 1 point - The company develops its own publicly accessible sustainability standards (environmental and/or social) that go beyond the legal requirements
- 1 point - The company involves external stakeholders in the development of sustainability standards (e.g. trade unions, NGOs, companies, government institutions)
- 1 point - The company is an active member of an initiative that aims to develop publicly accessible industry-wide or cross-industry sustainability standards.

Figure 14: Indicator “Engagement for the development of sustainability standards”

c) Indicator: Compliance with sustainability standards

As a responsible company, Nager IT aligns itself with the International Labour Organization (ILO) labour standards and designates them as target minimum standards in the supply chain, although they are not universally applied throughout the supply chain and are not fully implemented in the sub-supply chain under consideration.

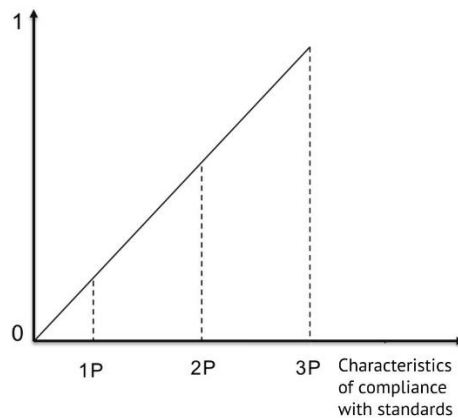
Additional essential standards are applied by key companies in the supply chain. For example, Stannol, as the manufacturer of solder wire, is a certified user of an environmental management system according to ISO 14001, and metal suppliers have presented “Conflict-free” certificates.

For the indicator, an “all or nothing” approach was designed for the evaluated group of companies, meaning either a standard is applied universally or not at all. Considering only the part of the supply chain for solder wire, adherence to ILO core labour standards as a social standard would therefore result in only some of the companies receiving points. The same applies to the other mentioned standards. However, individual criteria of the ILO core labour standards can be captured through other indicators (e.g., average wage).

This observation holds true regardless of the substitution process.

Compliance with sustainability standards

SDG-REFERENCES: 12.6, 12.8, 16.6.



Points system proposal

- 0 points – No compliance with standards
- 1 point - Compliance with an environmental or social standard
- 1 point - Compliance with a comprehensive sustainability standard or an environmental and social standard
- 1 point - Independent external audit of at least one standard used

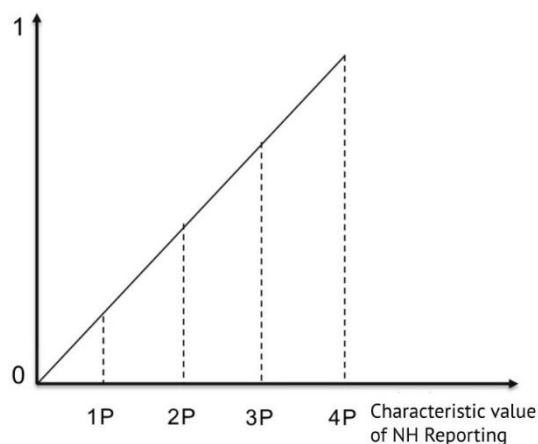
Figure 15: Indicator “Compliance with sustainability standards”

d) Indicator: Integration of sustainability information into reporting

Nager IT is formally registered as an association. The association provides comprehensive reporting on sustainability activities and, as part of this, offers detailed information (e.g., data sheets for the mouse). However, it does not adhere to a traditional reporting standard. Therefore, the indicator would be rated as “0” or might not even be considered in a preceding materiality analysis.

Integration of sustainability information into reporting

SDG-REFERENCES: 12.6., 12.8.



Points system proposal

- 0 points – No sustainability reporting
- 1 point - Sustainability reporting without standard orientation
- 1 point - Sustainability reporting, partly based on a relevant standard*
- 1 point - Sustainability reporting that explicitly follows a relevant standard*
- 1 point - Sustainability reporting that explicitly follows a relevant standard* and has an assurance statement

*e.g. GRI or comparable national standard

Figure 16: Indicator “Integration of sustainability information into reporting”

e) Indicator: Active communication of sustainability aspects to users

Nager IT communicates extensively and in a differentiated manner about a variety of relevant sustainability aspects through various channels (website, product), earning the company several points. The points would also be awarded even if the substitution did not take place: the indicator does not differentiate between "good" or "bad" but assesses relevance, as if something is noteworthy from a sustainability perspective. Therefore, the result for Nager IT is independent of the substitution process.

However, it would be worthwhile to question how the indicator could measure communication if a company communicated some but not all relevant aspects. In this case, the indicator does not provide a differentiation opportunity.

Active communication of sustainability aspects to users

SDG-REFERENCES: 12.8.

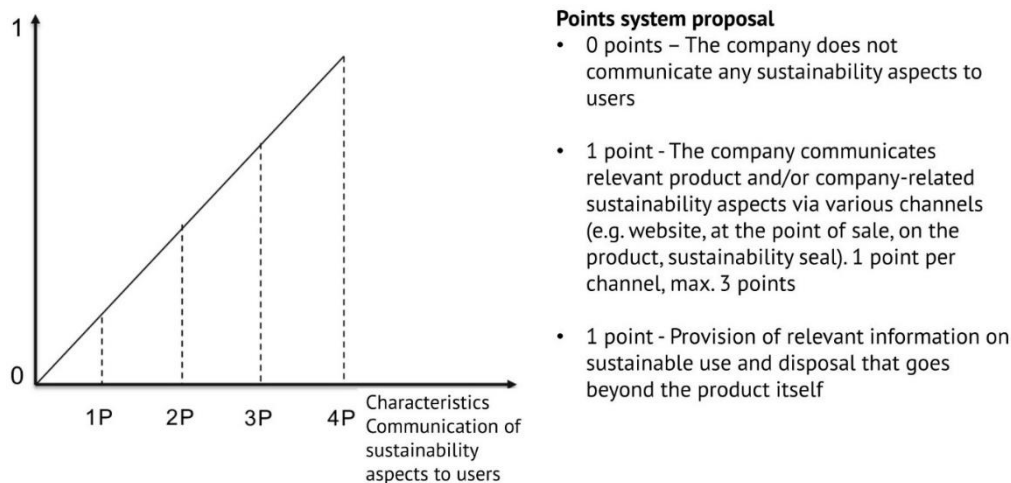


Figure 17: Indicator "Active communication of sustainability aspects to users"

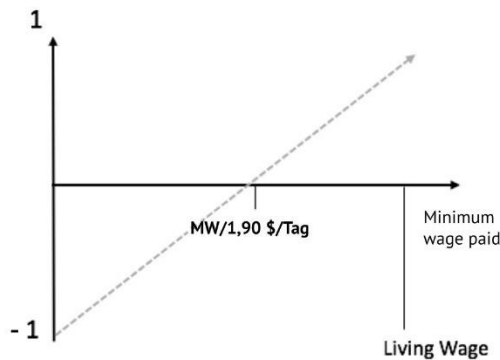
f) Indicator: Average wage of workers compared to minimum wage and existence minimum

Nager IT explicitly names the assurance of "dignified working conditions" as a goal, which also includes fair salary. Undoubtedly, the assurance of dignified working conditions is of high importance, especially at the upper end of the value chain in the extraction of metal raw materials, as illustrated. Even though, within the scope of the case study, no individual-specific data foundation could be created for comparing with a conventional mouse, it can be assumed that the substitution process and the use of recycled materials from Germany can potentially avoid social risks, including insufficient salary, in the extraction of metal raw materials. In conclusion, it can be stated that the

indicator is relevant for the case study, and the substitution process potentially generates a Handprint.

Average wage of workers compared to minimum wage and subsistence level

SDG-REFERENCES: 1.1., 8.5.



The following applies to all potential field curves:

Y-axis 0 to 1 Handprint range
Y-axis below 0 Footprint range

Y-axis 1 means 100% and thus the greatest possible contribution to the SDG.

Remark:

Minimum wage (MW) per day applies if this is higher than 1.90 \$ per day (UN minimum rate).

Figure 18: Indicator “Average wage of workers compared to minimum wage and existence minimum”

Analogously, this argumentation of avoiding Footprints through the substitution process can be applied to the indicators “**Ratio of male to female wages (in different salary levels)**” and “**Number of cases of child labour and forced labour**”. Additionally, it can be noted that the evaluation of standards such as compliance with ILO core labour standards through the Compliance with sustainability standards indicator may lead to double counting, as in the case of ILO core labour standards, the above indicators would implicitly result in a positive evaluation.

g) Indicator: Diversity and equal treatment promotion

For the upper end of the value chain, sufficient data could not be obtained for the application of the indicator. The indicator is detailed in case study 2. Since the origin of the material is irrelevant for assessing the working conditions during the soldering process conducted by Retex, it does not apply here.

Table 12 summarizes the results of the indicator analysis in the context of the case study on the substitution of tin by fair tin:

Table 12: Indicator analysis of case study substitution of solder tin

Indicator	Indicator result	Conclusion on methodology
Contribution to reducing waste	Handprint through material substitution	Applicability depends on proportion of relevant mass; should be checked by materiality analysis
Contribution to resource efficiency (efficient use of resources, reuse/recycling content, etc.)	No assessment possible as indicator not yet developed	
Commitment to the development of sustainability standards	Handprint through material substitution; further Handprints possible	Rewards committed companies only to a limited extent; if necessary, create the possibility of further weighting of activities
Compliance with sustainability standards	No Handprint, as indicator criteria insufficiently fulfilled	Examine whether (and how) it makes sense (and is possible) to award points for the application of standards only to parts of the supply chain
Integration of sustainability information into reporting	No Handprint, as indicator criteria insufficiently fulfilled	Examine whether practical application of the indicator as part of the materiality analysis can be made dependent on the legal form of an organization, for example
Active communication of sustainability aspects to users	Handprint independent of material substitution	Check the practical application of the indicator (how would the communication of partially relevant aspects be assessed?)
Average wage of workers compared to minimum wage and living wage	Handprint through material substitution	Risk of double evaluation by evaluating the aspect also by indicator Compliance with standards
Expenditure on social security	No assessment possible as indicator not yet developed	
Ratio of men's to women's pay (at different salary levels)	Handprint (reduction of Footprint) through material substitution	Risk of double scoring by evaluating the aspect also by indicator Compliance with standards
Number of cases of child labour along the product life cycle and number of cases of	Reducing the Footprint through material substitution	Risk of double scoring by evaluating the aspect also by indicator <i>Compliance with standards</i>

Indicator	Indicator result	Conclusion on methodology
forced labour along the product life cycle		
Promoting diversity and equal treatment	Insufficient data basis; Handprint for individual companies	see Diversity and equal treatment case study
Sustainability (risk) management in the company and in the supply chain	No assessment possible as indicator not yet developed	

Based on the indicator discussion, the following results can be noted:

Capture and Evaluation of relevant sustainability effects

Initially, it becomes evident that, fundamentally, all elaborated indicators can be applied to capture and assess sustainability effects discussed in the context of the substitution process. Some indicators would likely be excluded in practical application under the aspect of significance.

Furthermore, it is evident that the indicators can provide a nuanced picture in the sense that most Handprints were directly generated by the substitution process, while other effects also persist independently of the substitution process. In terms of deriving context-specific recommendations for a company, this aspect is significant.

Regarding the lack or insufficient capturing and evaluation of sustainability effects, the analysis reveals the following results: Several effects result in a zero rating, such as the application of standards in only a part of the supply chain or engagement limited to just one initiative. It would need to be examined to what extent these generally desirable sustainability effects could also be represented.

Due to the low mass share, the contribution to waste reduction is negligible. Consequently, a critical question arises regarding the significance of other sustainability effects, which, in a product-oriented perspective, may also pertain to only a small portion of the product. For instance, Nager IT's commitment within the framework of "Fairlötet". Should this commitment be more highly valued if the mass share of solder was 10% or 50%? A counterargument could be that the commitment must be viewed at the company level and is desirable under sustainability considerations, as defined by the Handprint methodology.

The discussion on working conditions could only be conducted superficially due to the limited dataset. However, it is notable that, based on the example of ILO core labour standards, separate

indicators are available for some criteria (salary) but not for others or only in a limited capacity (regulated working hours, health protection).

The fact that some effects can be represented across multiple indicators poses the risk of double counting (e.g., ILO core labour standards through adherence to standards and average wage). It could be explored to what extent individual indicators could be replaced by meta-indicators (e.g., adherence to standards, implementation of bundled measures, or similar).

2.4.2.2 Case study on diversity and equal treatment

Diversity is a significant topic in the management of many companies. A heterogeneous and diverse workforce is considered an asset for corporate culture and, ultimately, for productivity.

Workforce diversity encompasses a range of characteristics, including gender, age, origin, religion, worldview, sexual orientation, social background, and aspects of physicality such as disability. Treating individuals with these diverse attributes equally is not just a counter concept to discrimination but can be a tangible asset for work and economic relationships.

The United Nations has enshrined the promotion of diversity and equal treatment in SDG Sub-Goal 8.5: "By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value."

For this case study, we will examine Retex Werkstatt GmbH, whose only shareholder is the association "retex – Regensburger Initiative zur Schaffung von Arbeitsplätzen für psychisch kranke und behinderte Menschen" (Regensburg Initiative for the Creation of Jobs for Mentally Ill and Disabled People). Founded in 1985, the association aims to create employment opportunities for mentally ill, disabled, and long-term unemployed individuals. The workshop's integration work has received multiple awards and is considered exemplary. The assembly of computer mouse components and the packaging of computer mice take place in the workshop, carried out by mentally ill and disabled individuals. The workstations are tailored to individual needs, offering flexible working hours, rest areas, and personal support. Targeted qualifications are provided to facilitate the transition to the regular job market.

Through this case study, we aim to determine to what extent the methodology can adequately and distinctly evaluate the measures chosen by Retex Werkstatt to integrate individuals with disabilities. The goal is to provide companies and end consumers with helpful information for aligning their actions with sustainability. Specifically, we will examine whether the methodology can capture these measures and, in its evaluation results, practically serve as a decision-making aid for companies and end consumers in favour of more sustainable practices.

The Handprint methodology provides the following indicator to measure and evaluate the *Integration of People with Disabilities* in an organization:

Promotion of Diversity and Equal Treatment

This indicator is related to the SDG Sub-Goal 8.5.

The following two examination questions will be explored:

- Accuracy and Practicality: To what extent can the integration measures taken in practice, using the example of Retex Werkstatt, which are assumed to be useful a priori due to the nature of Retex Werkstatt, be reflected through the indicator?
- Action Orientation: Does the assessment using the indicator result in an action-oriented approach tailored to the respective stakeholders, meaning, are the results usable for companies and end consumers in a way that provides opportunities for more sustainable actions?

Introduction of the indicator

The presented indicator and the underlying potential field curve are preliminary versions that have been developed in the project through the multi-method approach up to this point but have not been conclusively determined or verified by external stakeholders.

Promoting diversity and equal treatment

SDG-REFERENCES: 8.5. ACHIEVE FULL AND PRODUCTIVE EMPLOYMENT AND DECENT WORK FOR ALL WOMEN AND MEN, INCLUDING FOR YOUNG PEOPLE AND PERSONS WITH DISABILITIES, AND EQUAL PAY FOR WORK OF EQUAL VALUE

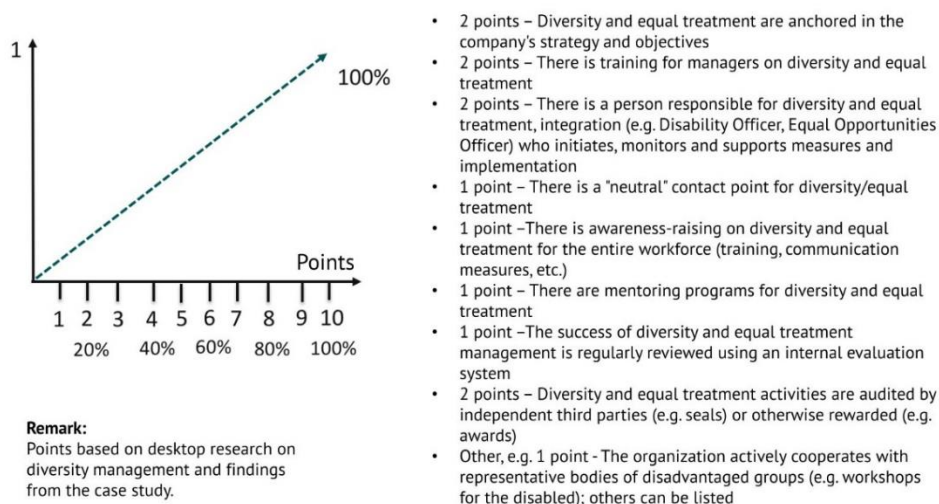


Figure 19: Indicator "Promotion of diversity and equal treatment"

Examination of Accuracy and Practicality

It will be examined to what extent the integration measures carried out in practice, using the example of Retex Workshop, which are assumed to be useful a priori, are adequately represented through the indicator. Are there important measures taken by Retex that are not taken into account? Does the indicator, conversely, include important criteria that cannot be fulfilled?

The investigation of Retex Workshop¹² has shown that the following measures for the integration of mentally ill and disabled people are applied in practice:

- The creation of jobs and employment opportunities for mentally ill and disabled people is an inherent concern of the company and strategically anchored.
- Two managers dedicate approximately 3 and 3-5 hours per week, respectively, to current sustainability topics (environment and social issues).
- Internal training sessions for group leaders on sustainability topics, including the use of pictograms; additional training sessions within a Continuous Improvement Process in case of "failures of measures"; engagement of an external quality officer for 10 hours per month
- Apprentices are each mentored by a designated instructor
- Regular "social competence training" for employees
- Collaboration with two workshops for people with disabilities, involving knowledge and experience exchange as well as the placement of workers
- Collaboration with social workers from the youth welfare office and acceptance of hard-to-place youths for training (although, in previous cases, some youths did not complete their training as they dropped out voluntarily)
- Participation in the career orientation event "Girls' Day" to showcase technical professions
- Close cooperation with other educational programs and initiatives (Youth Technology House, "Curiosity Express" for early awareness-building for technical professions)

A comparison with the indicator criteria reveals the following picture:

Table 13: Evaluation of Retex workshop measures

Indicator criterion	Measures Retex	Rating	Comment
Diversity and equal treatment are anchored in the company's strategy and objectives	Creating jobs and employment opportunities for mentally ill and disabled people is a fundamental concern of the company and is strategically anchored	2 points	

¹² Data collection via interviews and an on-site visit.

Indicator criterion	Measures Retex	Rating	Comment
Training for managers on diversity and equal treatment	As part of training on sustainability topics in general (2 managers)	1 point out of a possible 2	Assumption that criterion is at least partially fulfilled by general NH orientation; room for interpretation
Responsible person for diversity and equal treatment, integration	No explicit representatives for the topic, but targeted promotion of disadvantaged persons as a fundamental concern of the company and part of the management strategy	2 points	Assumption that all managers are responsible implementers; room for interpretation
"Neutral" contact point for diversity / equal treatment	Lecturer as contact person, but not a "neutral" person	0 points	
Raising awareness of diversity and equal treatment for the entire workforce (training, communication measures, etc.)	General sustainability training at manager and group leader level and regular "social skills training" for employees	1 point	Assumption that criterion fulfilled; room for interpretation
Mentoring program for diversity and equal treatment	No explicit mentors for the topic across the board, but targeted support for disadvantaged people through lecturers	1 point	Assumption that criterion fulfilled; room for interpretation
Regular review of the success of diversity and equal treatment management through an internal evaluation system	Routine identification of failures of sustainability measures in general and then follow-up training as part of a continuous improvement process; the extent to which an internal evaluation system is used is questionable	0 points	Assumption that criterion not fulfilled as no evaluation system; questionable to what extent this is reasonable to expect for an SME
Diversity and equal treatment activities are audited by independent third parties (e.g. seals) or otherwise rewarded (e.g. awards)	Use of an external quality officer, also to identify failures of sustainability measures in general; multiple awards (including the Luise Kiesselbach Prize 2017) for social commitment in general	1 point	Assumption that criterion is also fulfilled by general NH measures; scope for interpretation
The organization actively	Several cooperation	1 point	

Indicator criterion	Measures Retex	Rating	Comment
cooperates with representative bodies of disadvantaged groups (e.g. workshops for the disabled)			

After this evaluation, Retex scores 9 points, fulfilling 90% of the indicator criteria.

Regarding the completeness question, it can be observed that initially, all the listed measures of the Retex Workshop, which were assumed to be useful and thus essential, are considered in the discussion of fulfilling the indicator criteria; some are considered multiple times (e.g., training measures), others just once (e.g., collaborations - only relevant for the last-mentioned indicator criterion). The question of criterion fulfilment often leaves room for interpretation: for instance, to what extent training measures on sustainability in general also address the specific aspect of diversity and equal treatment, or how the term "mentor" is understood. Interpretation margins could be narrowed down by specifying data collection (What are the exact contents of the training?), and providing interpretation guidelines with, for example, additional examples (who qualifies as a mentor?).

Formally and technically, it needs to be clarified whether a criterion with a point value of 2 can be considered partially fulfilled with 1 point or if an "all or nothing" rule should apply.

The indicator also specifies two criteria that were not fulfilled according to this assessment. However, discussions about criterion fulfilment show that Retex, with its measures, at least offers something here, even if it doesn't lead to full criterion fulfilment.

The question of the completeness of the indicator criteria using an example of a company that can be considered almost exemplary in the field of diversity and equal treatment shows a high level of correspondence between the offered measures and the criteria to be fulfilled: there is neither a measure that does not match at least one criterion, nor criteria that are not addressed by any measures.

Based on this case study, it can be concluded that the indicator encompasses essential criteria. However, further case studies need to be examined to assess the completeness of the evaluated criteria. It should be considered that in this case study, the promotion of diversity and equal treatment exclusively focuses on the integration of mentally ill and disabled people. Aspects such as gender ratio, sexual orientation, religious affiliation, or others were not considered.

Implementation Guidance

It is intended to examine to what extent the method of assessing the promotion of diversity and equal treatment provides results that are useful for both companies and end consumers, offering them safer options for more sustainable actions.

As the reuse of results (e.g., interaction with other SDG 8-related results, presentation) within the framework of the project could not be completed using the Handprint methodology, this analysis is severely limited.

A detailed analysis of the results, as shown above, could give companies clear indications on how contributions to SDG sub-goal 8.5, the promotion of diversity and equal treatment, can be increased through specific measures. This knowledge supports management decisions and thus fulfils a fundamental expectation of the methodology (see Working Paper 2). However, it does not show, for example, at this point, what conflicts of interest may be associated with these measures.

For **end consumers**, the information generated could indicate how well a company is performing in the area of “Promotion of Diversity and Equal Treatment” concerning the achievement of SDG 8.5. Depending on the consumer’s preference, they can reward or reject a specific contribution of the company through their purchasing decision. However, results cannot be provided that infer the integration of specific groups of people (e.g., religious minorities, individuals with specific sexual orientations). In this context, an extension of the set of criteria to generate information in this regard would be desirable.

Result

Based on the case study of Retex, it can be inferred that the indicator “*Promotion of diversity and equal treatment*” is based on essential criteria, i.e., criteria that are useful and meaningful in practice.

However, these findings should be validated through additional case studies and in different contexts to establish a larger database. It should be noted that in this case study, the promotion of diversity and equal treatment was exclusively examined regarding the integration of individuals with mental illnesses and disabilities. Aspects such as gender ratios, sexual orientation, religious affiliation, and others were not considered.

Further insights reveal that there are interpretation gaps in the application of the indicator, which can ultimately lead to a subjective assessment. Some remedies could include a more precise data collection process and a supportive application guide with interpretation aids, such as examples.

Regarding recommendations, as an initial result, it can be noted that the indicator initially allows clear implications for companies to take sustainability-promoting measures based on the promotion of diversity and equal treatment. For end consumers, the results could provide a good orientation regarding the discussed criterion of diversity and equal treatment (with appropriate presentation). However, it does not allow decision-making regarding specific groups of people, such as religious minorities or individuals with certain sexual orientations.

2.4.2.3 Case study of supply chain transparency

Understanding the structure and functioning of global value chains is a crucial prerequisite for managing associated risks and opportunities in favour of sustainable development. Creating transparency in supply chains is, in turn, a vital requirement for generating such understanding.

The United Nations incorporates the idea of transparency as an enabling factor for sustainable development in the following SDG sub-goals:

SDG 12.8: *By 2030, ensure that people everywhere have the relevant information and awareness for sustainable development and lifestyles in harmony with nature.*

SDG 12.6: *Encourage companies, especially large and transnational companies, to adopt sustainable practices and to integrate sustainability information into their reporting cycle.*

SDG 16.6: *Develop effective, accountable, and transparent institutions at all levels.*

In the project, the following indicators were developed to assess product or company contributions to creating transparency in value chains or fulfilling the mentioned SDGs, and these will be tested in this case study:

- *Integration of sustainability information into reporting*
- *Compliance with sustainability standards*
- *Engagement in the creation of sustainability standards*
- *Active communication of sustainability aspects to users*

Specifically, the investigation aims to address the following:

- **Differentiation:** To what extent do the indicators measure different measures?
Redundancies of the same measures should be avoided to prevent an imbalance in the assessment, where a single measure earns points for multiple indicators, contradicting the principle of equal weighting of indicators. The method is expected to provide clear implications for companies by allowing sustainability impacts of individual measures to be assigned to specific indicators.
- **Completeness:** Do the given indicators capture all relevant activities, or are essential transparency-determining activities disregarded—in the case of the Fair Mouse, as well as potentially in other contexts?

For Nager IT, transparency in the value chain is a deliberate and actively pursued objective. The goal is to disclose the entire supply chain of the Fair Mouse, making it accessible to everyone primarily through the company's website.

Additionally, Nager IT communicates sustainability aspects of the product through newsletters, YouTube, providing datasheets, and on the product itself (or in the packaging). The company also

shares information about repair and disposal options for the mouse. Moreover, Nager IT collaborates with youth leaders to conduct workshops where children and teenagers discuss the challenges of a Fair Computer Mouse and have the opportunity to assemble a computer mouse themselves.

Nager IT does not adhere to a specific reporting standard such as GRI (Global Reporting Initiative). However, the company strives to establish fair working conditions throughout the entire supply chain, aligning with ILO (International Labour Organization) standards. This includes a focus on

- Regulated working hours
- Fair compensation
- Health protection
- Social security
- Freedom of assembly
- Exploitative child labour, forced labour, etc., are explicitly excluded

Nager IT envisions being a pioneer in promoting fair working conditions for all participants in global IT production.

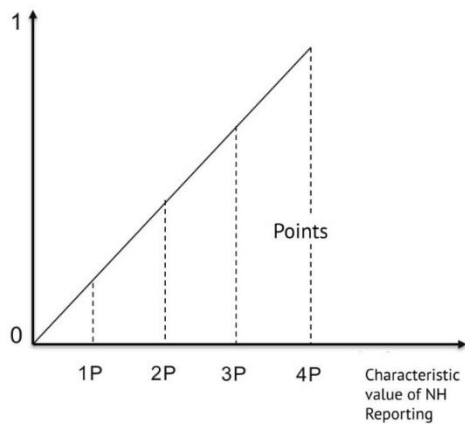
In the environmental domain, Nager IT does not follow specific standards like an environmental management system. Nevertheless, the company implements individual measures, such as the use of recycled solder, which is certified by "Fairlötet".

Introduction of indicators:

The indicator "*Integration of sustainability information into reporting*" evaluates the quality of sustainability reporting by classifying it based on criteria and assigning points. The assessment considers the alignment with existing reporting standards, the scope, and the reliability or credibility of compliance with these standards.

Integration of sustainability information into reporting

SDG-REFERENCES: 12.6., 12.8.



Points system proposal

- 0 points – No sustainability reporting
- 1 point – Sustainability reporting without orientation towards a standard
- 1 point – Nachhaltigkeitsreporting teilweise orientiert an einem relevanten Standard*
- 1 point – Sustainability reporting that explicitly complies with a relevant standard*
- 1 point – Sustainability reporting that explicitly complies with a relevant standard* and includes an assurance statement

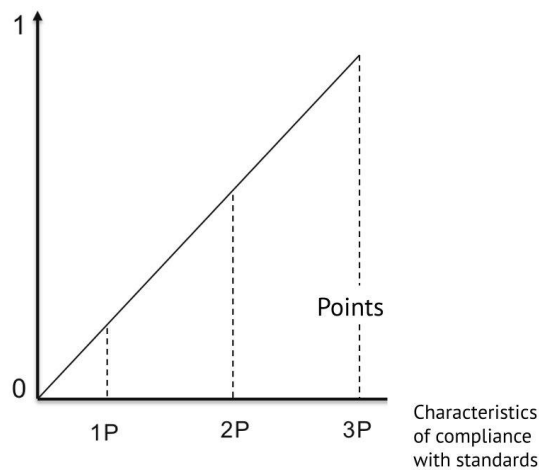
* e.g. GRI or comparable national standard

Figure 20: Indicator "Reporting"

The indicator "*Compliance with sustainability standards*" (Figure 21) captures the (product-relevant) applied environmental and social standards within a company. Ideally, a company would adhere to a standard in both areas (or an overarching sustainability standard spanning both areas) and have compliance verified through an independent external audit.

Compliance with sustainability standards

SDG-REFERENCES: 12.6., 12.8., 16.6.



Points system proposal

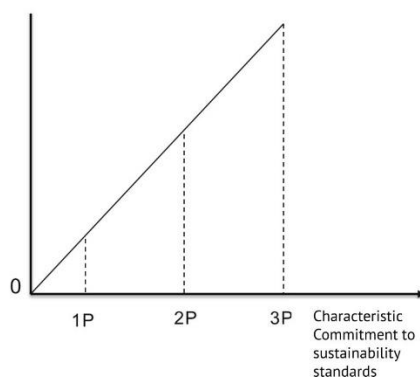
- 0 points – No compliance with standards
- 1 Point – Compliance with an environmental or social standard
- 1 Point – Compliance with a comprehensive sustainability standard or an environmental and social standard
- 1 Point – Independent external audit of at least one standard used

Figure 21: Indicator “Compliance with sustainability standards”

The indicator “*Commitment to the development of sustainability standards*” gauges the creation of new standards in the environmental and social domains, along with their quality, assessed through the criterion of involvement of independent third parties.

Commitment to the development of sustainability standards

SDG-REFERENCES: 16.6.



Points system proposal

- 0 points – No commitment to the development of sustainability standards
- 1 Point – The company develops its own publicly accessible sustainability standards (environmental and/or social) that go beyond the legal requirements
- 1 Point – The company involves external stakeholders in the development of sustainability standards (e.g. trade unions, NGOs, companies, government institutions)
- 1 Point – The company is an active member of an initiative that aims to develop publicly accessible industry-wide or cross-industry sustainability standards

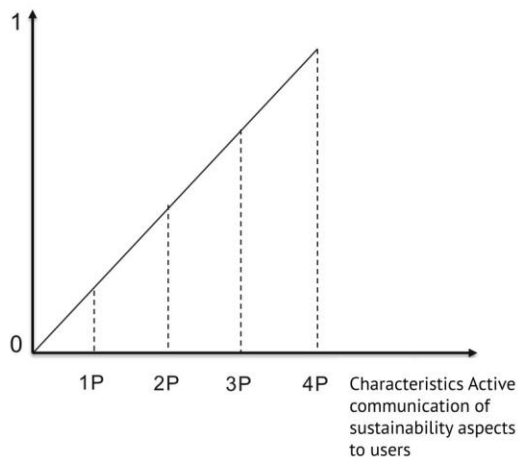
Figure 22: Indicator “Commitment to the development of sustainability standards”

The indicator “*Active communication of sustainability aspects to users*” is explicitly aimed at product users and assesses the number of communication channels through which relevant sustainability

aspects related to the product and/or company are communicated. It also evaluates the provision of information beyond the product, focusing on sustainable use and disposal.

Active communication of sustainability aspects to users

SDG-REFERENCES: 12.8.



Points system proposal

- 0 points – The company does not communicate any sustainability aspects to users
- The company communicates relevant product and/or company-related sustainability aspects via various channels (e.g. website, at the point of sale, on the product, sustainability seal). 1 point per channel, max. 3 points
- 1 Point – Provision of relevant information on sustainable use and disposal that goes beyond the product itself

Figure 23: Indicator “Active communication of sustainability aspects to users”

Analysis of Distinctiveness and Completeness

The analysis aims to examine to what extent the indicators are based on similar or different criteria and whether, in practice, the criteria assess similar measures. Additionally, it will assess how well the company's strategic and actively pursued goal of creating transparency can be reflected in the indicators.

The following overview provides a list of relevant measures taken by Nager IT and relates them to the indicators.

It is important to note that in the Handprint methodology, reporting standards and communication measures are classified under the governance category. This means they are not considered part of environmental or social standards.

Table 14: Creating transparency – test application of the indicators

Measures Nager IT e.V.	Indicators capturing measures			
	Integration of sustainability information into reporting	Compliance with sustainability standards	Commitment to sustainability standards	Communication
Creating transparency as a strategic goal	Yes: the associated package of measures can be seen as "sustainability reporting without orientation to standard (1/4 points)"	Not included in the criteria grid, as governance measure	Not included in the criteria grid, as a governance measure; however, disclosure of the value chain could be seen as a transparency standard	Not included as a strategic objective in the criteria grid
Fair working conditions as a strategic goal; practically in line with ILO core labour standards	Not a reporting/governance measure, but a social measure	Yes: ILO core labour standards relevant; however, no full application of the standard (0/3 points)	The company pursues its own requirements, which exceed the given standards in certain areas and thus aims to bring about change, but does not pursue the development of new standards in the narrower sense	Not included in the criteria grid, as social measure
Individual environmental sub-targets, no orientation towards standard	Not a reporting/governance measure, but a social measure	Yes: "FairLötet" certificate; however, no comprehensive environmental standard (0/3 points)	No commitment to the development of new environmental standards in the narrower sense	Not included in the criteria grid, as environmental measure

Communication of comprehensive NH information to the public via various channels	Yes: Communication measures could be seen as "sustainability reporting without orientation to standard (1/4 points)"	Not included in the criteria grid, as governance measure	Not included in the criteria grid, as governance measure	Yes: Measures are explicitly recorded using a criteria grid (4 points)
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The comparison shows that the analysed measures related to transparency creation, fair working conditions, and those related to the environment are captured by at most one indicator, thus providing a clear distinction between the indicators. Only communication measures would be captured by the indicators "*Reporting*" and "*communication to users*", posing the risk of double counting. If reporting is understood as purely internal, double counting could be avoided. In the case of reporting both internally and externally, different sets of criteria could be established for each or the indicator "*Communication to users*" could be expanded to include communication to the public.

The analysis yields further insights and implications for the methodology:

- Regarding **environmental and social standards**, it remains unclear how comprehensive a standard must be for fulfilment. Additional details or application examples would be helpful here.
- **Vertical coverage along the supply chain:** The assessment of Nager IT for the Compliance with Sustainability Standards indicator yields no points since Nager IT does not pursue environmental or social standards itself. Nager IT is only responsible for distribution and implementation. However, the situation is different in the supply chain, where companies hold certifications such as ISO 14001 (environmental management) or ISO 50001 (energy management). Additionally, FSC- and PEFC-certified companies are part of the supply chain. In the social realm, OHSAS 18001 and SA 8000 standards are present. The methodology should provide a rule for addressing this circumstance. It could be considered that an evaluation of compliance with environmental and social standards in the case of Nager IT as a distribution company might be excluded as a result of a materiality analysis, while engagement could be considered as a desirable positive impact.

Regarding the **creation of transparency** as a strategic and actively pursued goal of the company, the analysis yields the following result:

The strategic measure would be rated with only one point, even though it leads to a desirable and fundamental effect, namely enabling sustainability as whole, as stated at the beginning. This relatively low rating is not conducive and should be corrected. Expanding the scope of the respective

indicators “*Compliance with standards and development of standards*” to include governance standards could address this concern. Alternatively, transparency could be measured by a separate indicator, based on this case, for example, through a percentage of the disclosed supply chain.

Conclusion

The analysis of the four indicators regarding transparency creation leads to the following conclusions:

- The differentiation of the indicators is largely given based on the case study. The risk of double-counting communication measures can be avoided by adjusting the indicators.
- Regarding the evaluation of transparency creation, the application of the indicators results in an unsatisfactory outcome since the expected and desirable effects are not adequately captured as Handprints. Correction approaches could involve expanding the scope of existing indicators to include the governance domain or creating an additional indicator that could measure, for example, the degree of supply chain disclosure.

2.4.3 Building & living case studies

Author: Samantha Luisa Silva

The case study *Construction & Living* focuses on the SME Werkhaus as the main actor. Werkhaus sells various products in a “plug-in system,” including furnishings such as shelves or stools, which is the reason for selecting it as a case study partner in the Construction & Living sector. In addition, Werkhaus offers variable display systems in the business customer area.

The best-selling item is the smaller piece of furniture shaped like the iconic “VW T1 Bus” (see Figure 24) designed for storing pens and office supplies. The product is made of medium-density fibreboard (MDF) sourced from initial thinning (conscious thinning during the care of managed forests), with a focus on the lowest emission classes (E0 and E1-quality). The boards are precisely cut with a laser. These boards can then be printed or otherwise decorated, such as colouring. Colorations are created with low-emission and environmentally friendly colours. Residual colours are reused. For surfaces subjected to intensive use, a UV protective lacquer is sometimes applied. The finished boards are then assembled and secured with rubber bands.



Figure 24: Model “VW T1 Bus”

These rubber bands are made of ethene-propene-diene-monomer (EPDM), a synthetic rubber. Care is taken to ensure it does not contain toxic plasticizers and is environmentally friendly.

Data collection is based on publicly available information and selected input through inquiries to the company. The following graphic vividly illustrates the product's supply chain and focuses on the supply chain section within the scope of the case study.

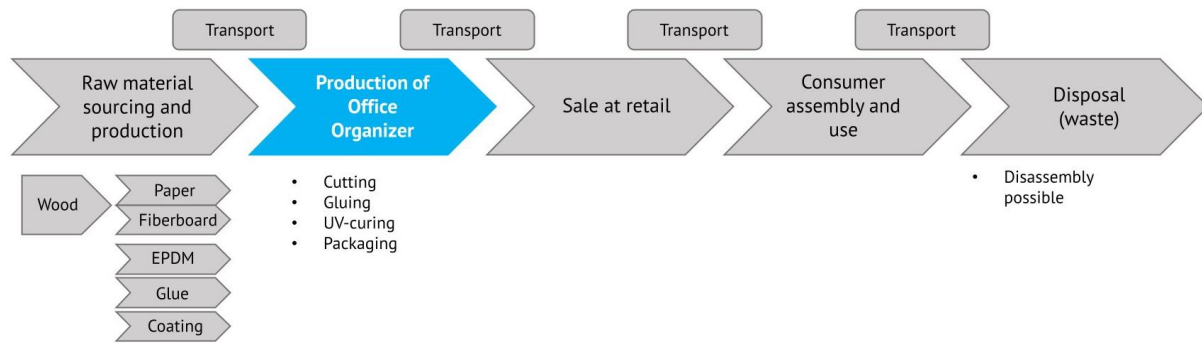


Figure 25: Schematic supply chain for the “VW T1 Bus” product at Werkhaus

While there is no alternative product considered for the identified product within the Handprint, it is evident that other pen holders are generally not produced in Germany and may contain other, less sustainable components such as plastic, hardwood, or metal. In most cases, manufacturers do not provide consumers with sustainability information, as is the case with Werkhaus.

In the selected case studies, two indicators are particularly discussed: “*Contribution to waste reduction*” (environmental category) and “*Transparency and standards at the company and product levels*” (governance category). These were examined as contributing indicators to *SDG 12, “Ensure sustainable consumption and production patterns”*, in two individual case studies. Table 155 provides an overview of the selected sub-goals with relevant indicators. This led to Case Study 3, focusing on transformational impact. The results and insights from the case studies are presented and discussed below.

Table 15: Sub-goals and selected indicators for SDG 12 – “Ensure sustainable consumption and production patterns”

SDG Sub-goal	Indicator	Range
12.4 By 2020, achieve the environmentally sound management of chemicals and all wastes throughout their life cycle, in accordance with agreed international frameworks, and significantly reduce their release to air, water and soil in order to minimize their adverse impacts on human health and the environment	Contribution to reducing waste	Environment

12.5 By 2030, substantially reduce waste generation through prevention, reduction, recycling and reuse.	Contribution to reducing waste	Environment
12.6 Encourage companies, especially large and transnational companies, to adopt sustainable practices and to integrate sustainability information into their reporting cycle.	Transparency and standards at company and product level	Governance
12.8 By 2030, ensure that people everywhere have the relevant information and awareness for sustainable development and lifestyles in harmony with nature.	Transparency and standards at company and product level	Governance

2.4.3.1 Case study SDG 12 – Ensure sustainable consumption and production patterns (focus on environmental indicator)

The indicator “*Contribution to waste reduction*” primarily considers the proportion of recycled materials per ton of processed materials. Werkhaus reports this proportion as 82.74%. The calculation involves determining the amount of recycled or reused material in relation to the total weight of the product. Additional information for this indicator includes Werkhaus’ general commitment to reducing waste during production, such as the reuse of unused paint.

Table 16: Results of selected indicators for SDG 12 – “Ensure sustainable consumption and production patterns” (focus: environmental indicator)

Indicator	Result	Additional information
Contribution to reducing waste	Proportion of recycled materials per ton of processed materials: 82.74%	- Commitment to minimize waste during production - Delivering products for consumer assembly reduces volume, thus also reduces packaging materials and shipping energy - Materials can be disassembled after use phase - No data on share of returned materials - Amount of waste by type per thousand Euros of revenues

- Amount of toxic waste per employee
- Waste is sorted and recycled
- Using wood waste for heating

To assess the contribution to the SDG, potential field curves are utilized. For the calculated indicator, a linear increase is assumed, with the maximum desired contribution being 100%, corresponding to a contribution of 1 (within the developed methodology). Therefore, at Werkhaus, with a share of 82.74%, the calculated contribution to the SDG is 0.83. The potential field curve is illustrated below:

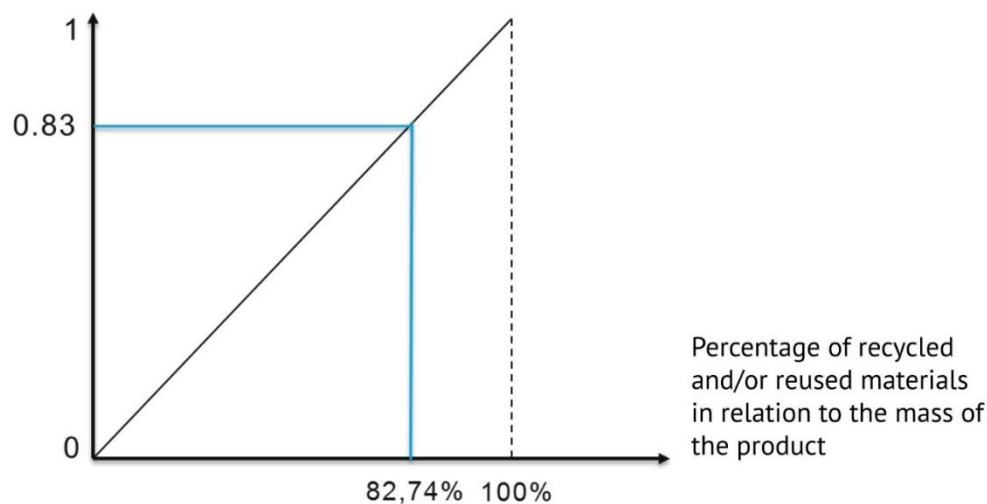


Figure 26: Potential field curve "Contribution to waste reduction/amount of waste"

Based on the developed evaluation, the summary interpretation of the relevant targets of SDG 12, namely 12.4 and 12.5, is as follows: It becomes clear that determining the contribution for SDG 12.4 requires further development and specification within the framework of the Handprint, and a better indicator specification would be helpful to make a clear statement. Developing a potential field curve or providing a clear definition of chemical substances could be possible in this regard. In the production description, Werkhaus makes it clear that this is an important detail of their production understanding, but it cannot currently be reflected in the Handprint.

For SDG 12.5, Werkhaus makes a significant contribution, as it achieved 0.83 out of 1. The interpretation follows the scale: 1 = very high contribution, higher than 0.67 high contribution, higher than 0.33 medium contribution; less than 0.33 low contribution, and 0 corresponds to no

contribution. Thus, based on the indicator proposed by the Handprint, Werkhaus makes a high contribution to SDG 12.5. Table 17 provides an overview:

Table 17: Targets and selected indicators for SDG 12 – “Ensure sustainable consumption and production patterns” (focus: environmental indicator)

SDG Sub-goal	Indicator	Contribution to the SDGs
12.4 By 2020, achieve the environmentally sound management of chemicals and all wastes throughout their life cycle, in accordance with agreed international frameworks, and significantly reduce their release to air, water and soil in order to minimize their adverse impacts on human health and the environment	Contribution to reducing waste	No specific indicator provided by Handprint approach in addition for the indicator addressing goal 12.5.
12.5 By 2030, substantially reduce waste generation through prevention, reduction, recycling and reuse.	Contribution to reducing waste	High contribution: contribution score of at least .83 reached in 2015; more time to achieve a maximum aspiration level by 2030

While the selected environmental indicator is advantageous in reducing negative impacts on the environment, the positive effect may not be the primary focus. However, the indicator is informative in highlighting sustainable production patterns. The positive impact on sustainable consumption patterns is not immediately evident in this indicator.

2.4.3.2 Case study SDG 12 – “Ensure sustainable consumption and production patterns” (focus on governance indicator)

Following the indicator *Transparency and Standards at the Company and Product Level*, the Integration of Sustainability Information into Reporting is examined based on the developed method and potential field curves. For the indicator Transparency and Standards at the Company and Product Level, the use of the Global Reporting Initiative G4 guidelines was identified during the creation of the report. However, the report was not independently verified by a third party. Nevertheless, Werkhaus communicates additional information on sustainability at various points of sale.

Table 18: Results of the selected indicators for SDG 12 – “Ensure sustainable consumption and production patterns” (focus: governance indicator)

Indicator	Result	Additional information
Transparency and standards at company and product level	Integration of sustainability information into the reporting: Use of Global Reporting Initiative G4 guidelines in preparation, no further third-party assurance	- Second sustainability report published in 2016 - Consumers receive sustainability information with products and at point of sale

Following the developed potential field curve for 'Integration of Sustainability Information into Reporting,' the resulting picture is as follows:

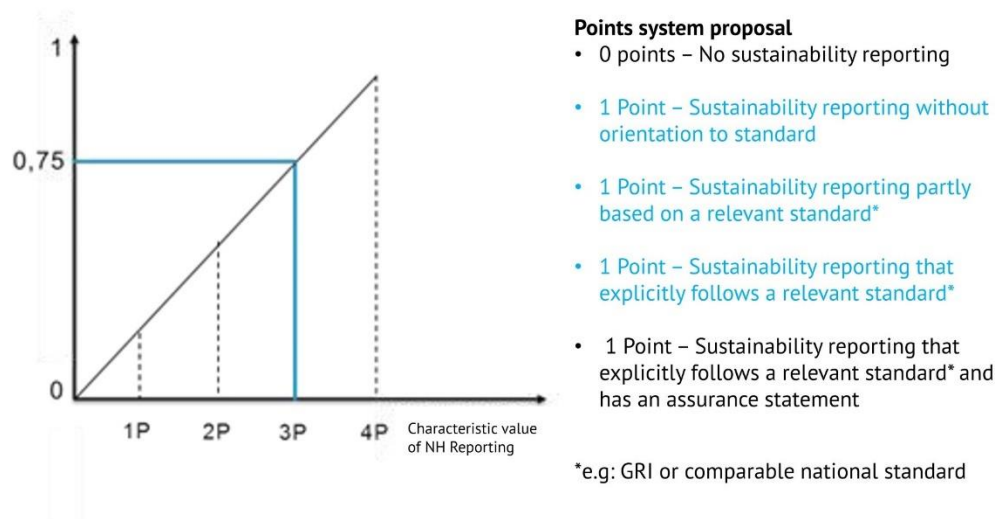


Figure 27: Potential field curve “Integration of sustainability information into reporting”

During the analysis, it became evident that the proposed point system is not initially clear and unambiguous: Does the system build upon each other, resulting in an increase across the stages, or does one receive one point per stage? Assuming the first two points are met, and a tiered system is established, Werkhaus would receive 3 points, equating to a contribution of 0.75 out of 1 and, following the previously introduced scale, representing a high contribution. illustrates the categorization. It is apparent that, even when breaking down into sub-goals again, 12.6 can be clearly justified, whereas 12.8 is not clearly assessable based on the indicator. With the identified additional information, it would be shown that Werkhaus conveys sustainability information at the point of sale. However, this cannot currently be captured in any specific indicator. Again, it becomes evident that further development and specification are required.

Table 19: Sub-goals and selected indicators for SDG 12 – “Ensure sustainable consumption and production patterns” (focus: governance indicator)

SDG Sub-goal	Indicator	Contribution to the SDGs
12.6 Encourage companies, especially large and transnational companies, to adopt sustainable practices and to integrate sustainability information into their reporting cycle.	Transparency and standards at company and product level	High contribution: contribution score of .75 reached in 2016
12.8 By 2030, ensure that people everywhere have the relevant information and awareness for sustainable development and lifestyles in harmony with nature.	Transparency and standards at company and product level	No specific indicator provided by Handprint approach in addition for the indicator addressing goal 12.6.

The selected governance indicator “*Transparency and standards at the company and product level*” shows potential effects on sustainable communication and production patterns based on additional information. At the point of sale, consumers are provided with information about the sustainability of the product. These details may influence consumers to choose this product, making them aware of the sustainability implications associated with their purchase decisions. This additional information could be an indication of changes in sustainable production patterns. However, whether this indicator would have identified this specificity in a large corporation is unclear and should be critically questioned and examined in the future application of the Handprint.

Certainly, the two selected indicators only shed light on a part of the indicators identified under SDG 12 in the Handprint; another one would be “*Engagement for the development of sustainability standards*”. However, the choice of these two case studies already exemplifies the difficulty of meaningfully representing SDG 12 and underscores the need for further development and specification.

While the idea and intention of SDG 12 focus on ensuring sustainable consumption and production patterns, applying it to a company reveals that the identified indicators, with a strong reliance on the life cycle, may present a distorted picture. The focus quickly orients towards the focal company and consumers, and the opportunities and changes of other market participants are easily forgotten. To identify a possible approach and further development, the transformation contribution was additionally addressed in one of the case studies.

2.4.3.3 Case study “Transformation impact”

Particularly, SDG 12, with a focus on ensuring sustainable consumption and production patterns, underscores the importance of change and, consequently, long-term transformation. This transformation indicates changes beyond company boundaries, such as alterations in suppliers and consumers, which have a positive impact in terms of sustainable development. In addition to examining and applying SDG 12 with its sub-indicators, an additional goal of the case study was to investigate the potential transformative effect of a small to medium-sized enterprise. For this purpose, not only was the individual product scrutinized, but also the selected case study company itself. Werkhaus GmbH + Co. (referred to as Werkhaus) was founded in 1992 by Eva Dannenberg, evolving from the Kaleidoskop Company. The company manufactures its products in Germany and sells them through its own online shop, additional online sales channels, as well as stores in the shopping passages of Berlin, Braunschweig, Hannover, Hamburg, Lüneburg, and Uelzen. It becomes evident that this is not a large corporation but a sustainability-focused enterprise that occupies a niche, making the examination of the transformation contribution challenging. To explore this, examples from other industries were drawn upon at various points, including the large corporation Unilever, as well as the SMEs Lichtblick and Bionade.

The case study understands transformational impact not only as processes of change within the focus company but also as changes extending beyond the company's boundaries. This includes examining changes that the case study company has exerted on suppliers, customers, and possibly competitors in the past and intends to do so in the future. Additionally, changes in market conditions are considered, including standards and legal alterations. Social impacts affecting the company's activities and products locally, regionally, or globally, are also taken into account. Lastly, environmental impacts have been considered. All these changes were contemplated using the categories from the Theory of Planned Behaviour¹³ by Icek Ajzen. The following overview presents a possible investigative approach:

¹³ Ajzen, I. (1991). The theory of planned behavior. *Theories of Cognitive Self-Regulation*, 50(2), 179–211

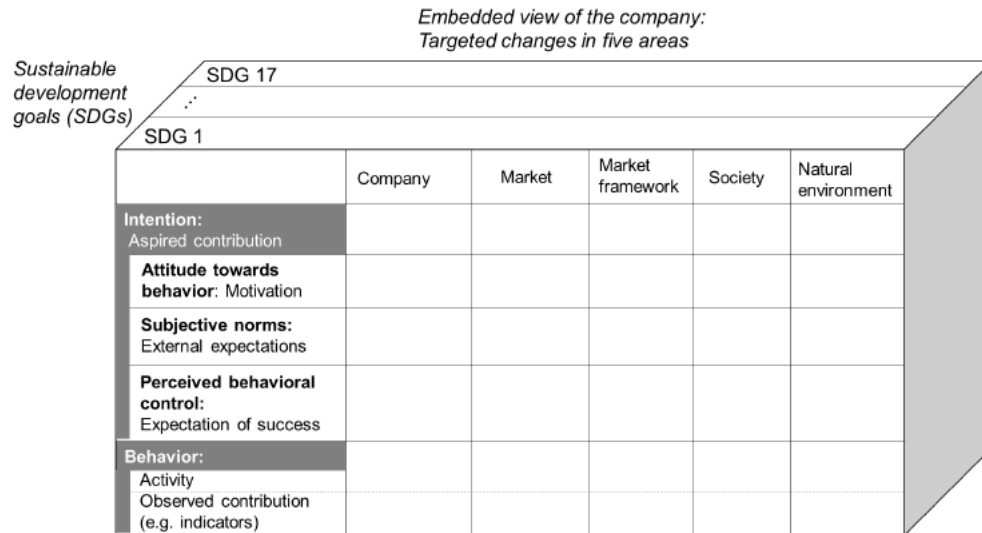


Figure 28: Possible categorization and examination of the transformational contribution (transformation cube)

When applying the categorization of the transformational contribution (hereinafter referred to as the Transformation Cube) to Werkhaus, it becomes evident that profound changes beyond the company's boundaries are challenging to identify. Nevertheless, the company's philosophy serves as a relevant starting point that could potentially lead to more significant transformations in the future. Examining Unilever as a large corporation through examples like the "Sustainable Living Plan" or the certification of the "Marine Stewardship Council" makes it clearer how a company can impact the market, market conditions, society, and the environment. This potential is also applicable to SMEs. For instance, Lichtblick experienced changes in laws regarding network fees, and Bionade faced a new legal categorization for beverages. These examples retrospectively illustrate market framework changes that resulted in more sustainable production and consumption patterns. The proposed Transformation Cube offers initial considerations on how such cross-company sustainable transformations can be envisaged from the outset.

The Handprint currently does not depict these transformations, but it illustrates the potential changes in management, measurement, and evaluation that could have broader implications. This undoubtedly needs further exploration, testing, and implementation in practice. However, it indicates future paths of development for positive contributions to sustainable development and emphasizes the necessity of further evolving the Handprint in the future.

The following **milestones** were achieved in AP2:

- M4: Completed execution and short report on the case studies; submitted on December 6, 2017

- M5: Publication of the final methodology and toolbox in the specialist journal UmweltWirtschaftsForum / NachhaltigkeitsmanagementForum (current revision status: minor revisions; initially submitted on December 18, 2017)

2.5 Work Package 3: Networking, dissemination, and multiplication

Work Package 3 aimed to facilitate the communication and dissemination of (interim) results. Its core objective was to establish a network through which the insights could be disseminated, reaching additional companies and various societal stakeholders. In addition to disseminating the results, the network was intended to serve as a platform throughout the project, incorporating external expertise from both national and international sources, spanning across diverse societal groups such as academia, businesses, and civil society. This collaboration was vital for integrating external perspectives into the methodology development process and exploring practical applications of the method. The assessment of application areas particularly considered industries and contexts beyond those addressed in the case studies.

Approach:

The core of Work Package 3 was the establishment of a network that brings together participants from various existing SME networks of research partners and practical partners, as well as from other businesses, academia, civil society, and politics. The basis for network development was a series of workshops, in which a broad range of stakeholders were involved alongside the research partners. This included experts who were, for instance, consulted in the Delphi studies in Work Package 1, representatives of SMEs involved in the pilot projects in Work Package 2, as well as other relevant stakeholders from politics, administration, and civil society.

The following stakeholder workshops were conducted as part of the project (see also the brief documentation of the workshops Milestone M6):

Table 20: Stakeholder-Workshop

Nr.	Date / Place	Title / Format	Target group	Number of participants	Topic
1	05.11.2015 Berlin	Kick-off workshop	Cross-stakeholder	29	Presentation of the project; identification of stakeholder requirements and expectations

2	31.05.2016 Wuppertal	Special workshop as part of CSCP's 10th anniversary celebrations	Cross-stakeholder, international	32	Presentation of the project, discussion of areas of application of the methodology; focus on the company perspective
3	01.06.2016 Wuppertal	Second stakeholder workshop	Cross-stakeholder	50	Presentation of the preliminary methodology and critical reflection
4	23.09.2016 Lüneburg	Third stakeholder workshop as part of the EMAN conference	Focus on science, companies, international	ca. 20	Presentation of the preliminary methodology and critical reflection; focus on the application area of sustainability accounting
5	09.11.2017 Wuppertal	Expert workshop	Practice partners from business and sustainable public procurement	9	Critical reflection on the methodology using selected indicators

Furthermore, the following measures accompanied the development of the network or were used to communicate project results:

- Establishment of a project website (www.handabdruck) to present the project, publish publications, and announce events
- Social media: Creation of project accounts on Twitter, with 89 followers (as of June 2018), and Facebook, providing regular updates on project progress, events, and publications
- Print: Design and printing of project flyers (approximately 250) and info cards (approximately 250) for distribution at events
- Regular presentation of the Handprint at various events such as conferences or workshops involving different stakeholders, both nationally and internationally, by all project participants; by June 2018, presented at over 20 events (see also annual reports)
- Publication of results (see Chapter 3.1)

The above measures have generated and continue to generate a broad response in the form of inquiries from interested individuals about the Handprint, fostering the growth of the network. The inquiries come from all stakeholders or target groups and usually concern potential applications of

the methodology, invitations to presentations, or the contribution of acquired (project) expertise (e.g., in similar initiatives of the inquiring individuals).

The network comprises over 400 qualified contacts (as of June 2018)¹⁴.

Results:

An achieved outcome was the network itself, which continues to exist and is further utilized for dissemination and subsequent activities related to the Handprint. Throughout the project, it proved highly valuable, enabling targeted outreach for participation in Delphi studies, stakeholder workshops, and other activities.

The regular inquiries from diverse stakeholders at all project partners highlight the potential of the Handprint methodology to contribute to the desired societal change. The CSCP (Collaborating Centre on Sustainable Consumption and Production) regularly incorporates the Handprint methodology as a conceptual approach in multi-stakeholder dialogues and sustainability-oriented business models. While the methodology as a whole has not yet been applied due to its developmental status within the project, interactions with academia, industry, and civil society have demonstrated that the methodology can be applied across various sectors and offers various potential benefits, such as a sustainability assessment tool, a steering tool (e.g., in public procurement), and an inspiration tool in product development.

The exchange activities in the project included not only presentations and workshop contributions but also collaboration with related research projects. During the project, there were collaborations with other FONA projects (see 1.5.) as well as (temporary) collaborations with Harvard University (Sustainability and Health Initiative for NetPositive Enterprise "Handprint"), VTT Institute Finland (Finnish national project on carbon Handprint), and GIZ (Corporate Sustainability Handprint®).

The following **milestones** were achieved in AP3:

- M1: Creation of the website and production of project flyers (December 2015)
- M6: Completion of the workshops, along with a brief documentation (November 2017)

3 Conclusion

The project goal, which was to develop the Handprint methodology, test it in case studies, utilize the experiences from pilot projects for additional products and industries, raise awareness of the Handprint in science, politics, and civil society, and establish connections to relevant international research projects, has been clearly achieved by the project. However, this new narrative and the focus

¹⁴ If required, contacts can be contacted online via the Mailchimp program.

on positive contributions have identified additional research needs and the necessity to further develop the Handprint.

3.1 Conclusion on method development

The goal of Work Package 1 (WP1) was to conduct a comprehensive inventory of relevant existing assessment methods and measurement approaches for the ecological and social aspects, as well as the negative and positive impacts of products. The objectives of WP1 were fully achieved, and the results were extensively published, making them available to a broad audience (see also Chapters 2.4 and 3.1). Based on the results of the literature analysis, practical approaches, and Delphi studies, concrete indicators for the assessment categories were developed in internal project discussions to establish the Handprint methodology as part of a holistic sustainability analysis. By integrating positive aspects, the Handprint methodology can ultimately contribute to a more comprehensive evaluation of the sustainability impacts of products.

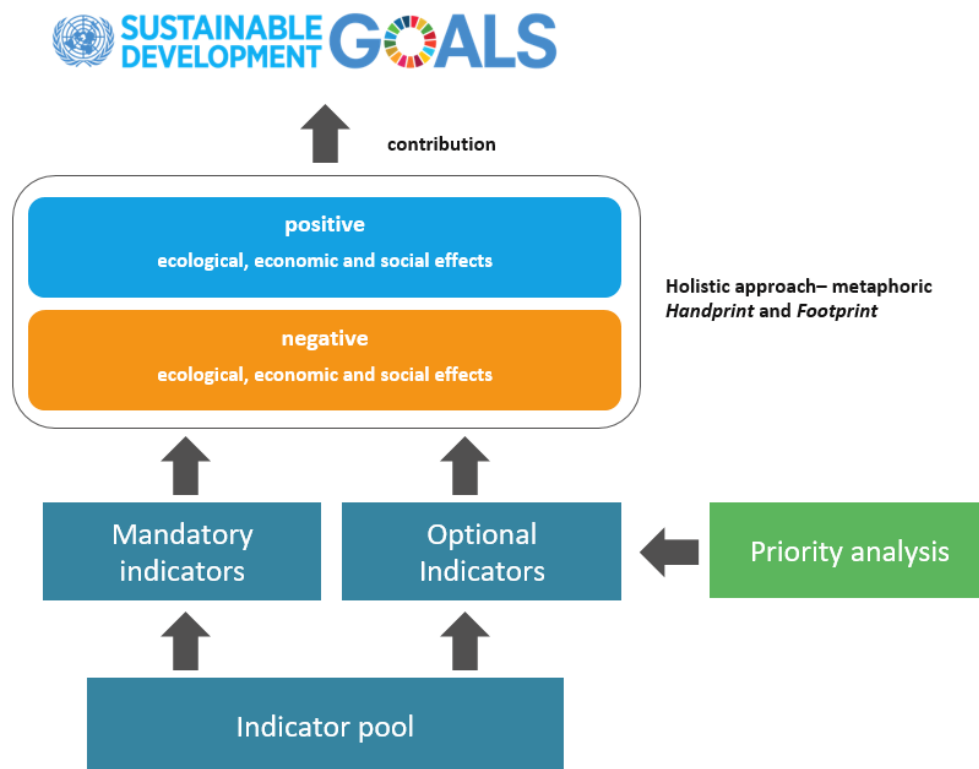


Figure 29: The methodological framework of the Handprint

3.2 Conclusion on case studies

In AP2, the following questions were examined through case studies:

- How well does the methodology developed in work package 1 perform in practice? Where is adaptation necessary for its application?
- Which sustainability effects or indicators to be examined are particularly relevant? Which indicators should be mandatory, and which optional? Does the relevance of indicators differ across the studied industries?
- What is the added value of the analysis for SMEs? How do companies specifically benefit from calculating the Handprint, and what incentives are there for companies to participate?
- What insights arise from practical implementation? What are the challenges and opportunities for improvement?

The evaluation in the food case study reveals that data retrieval takes a long time, as is common in life cycle assessments, and always longer than expected and agreed upon. Furthermore, data queries are never fully completed, resulting in data gaps that require a defined approach on how to handle them. This means that the methodology is consistently challenged to deliver precise results based on the highest possible specific primary data and, at the same time, to be practical.

The case studies in the consumer goods sector and around the *Fair Mouse* were oriented towards context-specific data collection and evaluation tests. It was demonstrated that positive sustainability effects (Handprints) can generally be precisely captured and assessed in practice using indicators. The resulting information can be useful for companies in terms of management decisions and for end consumers in terms of purchasing decisions. Further differentiation of the obtained results is considered desirable, especially regarding the *Diversity and Equal Treatment* indicator. The case study, conducted to assess the possibility of creating transparency about sustainability aspects in a supply chain, showed that the methodology offers this based on four indicators. There is potential for further development in the more precise alignment of indicator designs (and possibly the addition of further indicators) and to adjust the criteria for accurately measuring effectiveness.

The case study on building & living with the industry partner Werkhaus demonstrates that the selected SDG 12 (Ensure sustainable consumption and production patterns) can be represented in the developed Handprint methodology. However, an interesting area of consideration, the transformation beyond the company boundaries, is not systematically captured in the current version. A possible development, such as the representation in the Transformation Cube, illustrates the potential for a more extensive integration of this approach. Furthermore, the

Handprint approach currently serves for retrospective evaluation, and the implications for management can be derived subsequently. The Transformation Contribution case study highlighted the relevance of also discussing management implications and integrating the Handprint methodology to support management decisions.

In summary, regarding the initially posed questions:

The methodology provides the opportunity to systematically identify and measure Handprints. The use of SDGs as a benchmark for evaluation has proven effective. However, the proposed indicators and, in particular, the suggested evaluation still need further validation. This validation should ideally be done by stakeholders, which was only possible to a limited extent within the project.

It is still an open question how evaluation results regarding the SDGs can be further aggregated. Specifically, the methodological challenge of avoiding double counting needs to be addressed, along with proposing a procedure for handling conflicts between the SDGs.

The determination of mandatory and optional indicators could not be made based on the results of the case studies. However, the methodology envisages conducting a case-specific materiality analysis with the aim of identifying the relevant indicators in each case.

The Handprint method provides results that can be effectively utilized for management decisions, especially in product development, as it completes the assessment of negative impacts with the identification of positive effects, presenting a comprehensive picture. Additionally, it can offer new insights into a company's sustainability strategy. Areas for further development in decision support include expanding the scope of consideration (Transformation Cube).

As mentioned earlier, the results of the Handprint method complement those of Footprint methods. Together, they form the basis for the development of meaningful tools to convey sustainability information to consumers, providing guidance for consumer decisions. However, it is crucial to clarify which communication instruments are effective in promoting sustainable consumption patterns and how different target groups can be addressed with what means.

The data collection associated with the Handprint method is comparable in effort to other methods of product evaluation (e.g., Life Cycle Assessment). Similar to Life Cycle Assessments, the effort is relatively high for companies using the instrument for the first time. For companies that have already integrated product evaluation into their business processes and management decision-making – as is the case for some SMEs – the additional effort for implementing the Handprint is low. However, a fundamental challenge – similar to the early days of Life Cycle Assessments – is the lack of databases providing datasets for generic processes, such as the Handprint for transportation or energy generation. This will change with the increasing

acceptance of holistic sustainability assessment approaches, subsequently significantly reducing the effort required for method application.

Therefore, it is important to continue promoting the methodology beyond the project and emphasize the benefits of holistic sustainability assessment. The fundamental idea of expanding the sustainability perspective by incorporating positive effects resonates extremely positively across stakeholder groups (see also Section 3.3).

3.3 Conclusion on network formation, dissemination, and multiplication

Work package 3 was focused on the communication and dissemination of (interim) results, primarily aiming to establish a stakeholder network comprising academia, industry, and civil society. By the project's completion, the network included over 400 qualified contacts. Throughout the project, experts from the network were actively involved in the development work through surveys and workshops. Furthermore, the continued positive response to the communication of research results, as evidenced by ongoing inquiries into the methodology beyond the project's conclusion, underscores the lasting success of the project.

4 Literature References

- Ahi, P. & Searcy, C. (2015). Measuring social issues in sustainable supply chains. *Measuring Business Excellence*, 19(1), 33–45.
- Ajzen, I. (1991). The theory of planned behavior. *Theories of Cognitive Self-Regulation*, 50(2), 179–211
- Avilez, P. (2003). Bergbau in Chile und seine Auswirkungen auf die Umwelt. Download unter <https://www.tu-braunschweig.de/Medien-DB/geooekologie/exkursionsbericht-2003-pages189-200.pdf> (letzter Zugriff: 05.12.18)
- Antolín-López, R., Delgado-Ceballos, J. & Montiel, I. (2016). Deconstructing corporate sustainability: A comparison of different stakeholder metrics. *Journal of Cleaner Production*, Advance online publication.
- Arcese, G., Lucchetti, M. C., Massa, I. & Valente, C. (2016). State of the art in S-LCA: Integrating literature review and automatic text analysis. *International Journal of Life Cycle Assessment*.
- Baumann, H., Arvidsson, R., Tong, H. & Wang, Y. (2013). Does the production of an airbag injure more people than the airbag saves in traffic? *Journal of Industrial Ecology*, 17(4), 517–527.
- Benoît, C., Norris, G. A., Valdivia, S., Ciroth, A., Moberg, A., Bos, U., Beck, T. (2010). The guidelines for social life cycle assessment of products: Just in time! *International Journal of Life Cycle Assessment*, 15(2), 156–163.
- Beske-Janssen, P., Johnson, M. P. & Schaltegger, S. (2015). 20 years of performance measurement in sustainable supply chain management: What has been achieved? *Supply Chain Management: An International Journal*, 20(6), 664–680.
- Better Life Index: OECD. <http://www.oecdbetterlifeindex.org/de> (letzter Zugriff: 26.04.2017)
- Brandenburg, M., Govindan, K., Sarkis, J. & Seuring, S. (2014). Quantitative models for sustainable supply chain management: Developments and directions. *European Journal of Operational Research*, 233(2), 299–312.
- Bundesministerium für Bildung und Forschung (BMBF). (2013). *Nachhaltiges Wirtschaften: Sozial-ökologische Forschung*. [Online] Available at: <https://www.fona.de/de/nachhaltiges-wirtschaften-sozial-oekologische-forschung-17399.html>
- Bundesministerium für Bildung und Forschung (BMBF). (2016). *Forschung für Nachhaltige Entwicklung - FONAS*. [Online] Available at: https://www.bmbf.de/pub/Rahmenprogramm_FONA.pdf
- Burks, S. V., & Krupka, E. L. (2012). A multimethod approach to identifying norms and normative expectations within a corporate hierarchy: Evidence from the financial services industry. *Management Science*, 58(1), 203–217.

- Corporate Ecosystem Services Review: World Resources Institute.
<http://www.wri.org/publication/corporate-ecosystem-services-review> (letzter Zugriff: 27.04.2017)
- CSCP. (2016). handabdruck.org. Retrieved from Handabdruck: ein komplementäres Mass positiver Nachhaltigkeitswirkung von Produkten. Available at: handabdruck.org
- CSCP, CSM, Universität Hohenheim & ZNU. (2015). Handabdruck: Auftaktworkshop. Wuppertal: CSCP. Download unter: http://www.handabdruck.org/downloads/Handabdruck_Auftaktworkshop.pdf
- CSCP, CSM, Universität Hohenheim & ZNU. (2016). Handabdruck: 2. Stakeholder- Workshop. Wuppertal: CSCP. Download unter: http://www.handabdruck.org/downloads/Handprint_2nd_Workshop.pdf
- CSCP, CSM, Universität Hohenheim & ZNU. (2017). Handabdruck: Positive Nachhaltigkeitseffekte. URL: <http://handabdruck.org/index.php> (zuletzt geprüft am 30. Januar 2017).
- de Baan, L., Mutel, C.L., Curran, M., Hellweg, S., Köllner, T. (2013). Land Use in Life Cycle Assessment: Global Characterization Factors Based on Regional and Global Potential Species Extinction. *Environmental Science & Technology* 2013 47 (16), 9281-9290, DOI: 10.1021/es400592q
- Delmas, M. A., Etzion, D. & Nairn-Birch, N. (2013). Triangulating environmental performance: What do corporate social responsibility ratings really capture? *Academy of Management Perspectives*, 27(3), 255–267.
- DeNisi, A. & Smith, C. E. (2014). Performance appraisal, performance management, and firmlevel performance: A review, a proposed model, and new directions for future research. *Academy of Management Annals*, 8(1), 127–179.
- Denyer, D. & Tranfield, D. (2009). Producing a systematic review. In D. A. Buchanan & A. Bryman (Eds.), *The sage handbook of organizational research methods* (pp. 671–689). London: Sage.
- Di Cesare, S., Silveri, F., Sala, S. & Petti, L. (2016). Positive impacts in social life cycle assessment: State of the art and the way forward. *International Journal of Life Cycle Assessment*.
- Ekener, E., Hansson, J. & Gustavsson, M. (2016). Addressing positive impacts in social LCA: Discussing current and new approaches exemplified by the case of vehicle fuels. *International Journal of Life Cycle Assessment*, Advance online publication.
- Eberle, U., Hayn, D., Rehaag, R., Simshäuser, U., Stieß, I. & Waskow, F. (2005). Anforderungen an sozial-ökologische Vorsorge für eine Ernährungswende, Ernährungswende-Diskussionspapier Nr. 12, Download unter: www.ernaehrungswende.de
- Eberle, U.; Lindner, JP (2015). Biodiversity impact: Case study beef production; Proceedings of the „LCA for feeding the planet and energy for life“ conference 6-8 October 2015 in Stresa and Milan; pp. 302-306
- Ekener-Petersen, E. & Moberg, A. (2013). Potential hotspots identified by social LCA: Part 2 - Reflections on a study of a complex product. *The International Journal of Life Cycle Assessment*, 18(1), 144–154.
- Elkington, J. (1997). *Cannibals with forks: The triple bottom line of 21st century business*. Oxford: Capstone.

- FairLötet e.V. (2018). Warum besteht beim Lötzinn Handlungsbedarf? Hamburg
<https://fairloetet.de/wp-content/uploads/2018/10/Whitepaper.pdf>
- Fantke et al. (2015). USEtox® 2.0 User Manual (Version 2). <http://www.usetox.org>
- Feitz, A.; Dennien, G.; Morain, M.; Jones, M.; Lundie, S. (2005). Generation of an Industry-specific Physico-chemical Allocation Matrix. Application in the Dairy Industry and Implications for Systems Analysis (9 pp). Int J LCA 12 (2): 109–117.
- Feschet, P., Macombe, C., Garrabé, M., Loeillet, D., Saez, A. R. & Benhmad, F. (2013). Social impact assessment in LCA using the Preston pathway. International Journal of Life Cycle Assessment, 18(2), 490–503.
- Finnveden, G., Hauschild, M., Ekvall, T., Guinée, J., Heijungs, R., Hellweg, S., Suh, S. (2009). Recent developments in life cycle assessment. Journal of Environmental Management, 91(1), 1–21.
- Forum Informatikerinnen für Frieden und Gesellschaftliche Verantwortung e.V.: Faire Computer. https://www.fiff.de/themen/fair_it. Abruf: 09.01.2018.
- Gauthier, C. (2005). Measuring corporate social and environmental performance: The extended life-cycle assessment. Journal of Business Ethics, 59(1-2), 199–206.
- Global Social Venture Competition. (2014). Social impact assessment guidelines. Berkeley, US: Global Social Venture Competition; Haas School of Business.
- Global Social Venture Competition. (2015). Social impact assessment. URL: <http://gsvc.org/compete/how-to-enter/resources/social-impact-assessment/> (zuletzt geprüft am 11. Oktober 2015).
- Grießhammer, R., Buchert, M., Gensch, C.-O., Hochfeld, C., Manhart, A. & Rüdenauer, I. (2007). PROSA: Product sustainability assessment. Freiburg: Öko-Institut
- Hacking, T. & Guthrie, P. (2008). A framework for clarifying the meaning of Triple Bottom-Line, Integrated, and Sustainability Assessment. Environmental Impact Assessment Review, 28(2), 73–89.
- Hahn, R. (2011). Internationale Standardfindung und Global Governance: Zur Legitimität des Entstehungsprozesses der Leitlinie ISO 26000. Die Betriebswirtschaft, 71(2), 121–137.
- Hauschild, M. Z., Goedkoop, M., Guinée, J., Heijungs, R., Huijbregts, M., Joliet, O., Pant, R. (2013). Identifying best existing practice for characterization modeling in life cycle impact assessment. The International Journal of Life Cycle Assessment, 18(3), 683–697.
- Hunter, D. & Butz, S. (2009). Protocol for assessing investment risk with regard to the likelihood that an organization is producing social value. Hamden & Baltimore, US: Hunter Consulting; Social Solutions.
- Hütz-Adams, F. (2012). Von der Mine bis zum Konsumenten: Die Wertschöpfungskette von Mobiltelefonen. Siegburg, November 2012. http://www.suedwind-institut.de/fileadmin/fuerSuedwind/Publikationen/2012/2012-41_Von_der_Mine_bis_zum_Konsumenten_Die_Wertschoepfungskette_von_Mobiltelefonen.pdf. Abruf: 09.01.2018

- IDF (2010): A common carbon Footprint for dairy. The IDF guide to standard lifecycle assessment methodology for the dairy industry. Bulletin of the International Dairy Federation 445, 38 pp
- Lindner, JP (2016): Quantitative Darstellung der Wirkungen landnutzender Prozesse auf die Biodiversität in Ökobilanzen. Forschungsergebnisse aus der Bauphysik, Band 24, Fraunhofer Verlag, Stuttgart
- Iribarren, D., Martín-Gamboa, M., O'Mahony, T. & Dufour, J. (2016). Screening of socio-economic indicators for sustainability assessment: a combined life cycle assessment and data envelopment analysis approach. The International Journal of Life Cycle Assessment, 21(2), 202– 214.
- ISO. (2006). ISO 14040: Environmental management – Life cycle assessment – Principles and framework. Geneva: ISO.
- ISO. (2010). ISO 26000: Guidance on social responsibility. Geneva: ISO.
- Jeswani, H. K., Azapagic, A., Schepelmann, P. & Ritthoff, M. (2010). Options for broadening and deepening the LCA approaches. Journal of Cleaner Production, 18(2), 120–127.
- Jørgensen, A., Lai, L. C. & Hauschild, M. (2010). Assessing the validity of impact pathways for child labour and well-being in social life cycle assessment. International Journal of Life Cycle Assessment, 15(1), 5–16.
- Kabongo, J. D. & Boiral, O. (2017). Doing More with Less: Building Dynamic Capabilities for Eco-Efficiency. Business Strategy and the Environment, 10(3), 273.
- Klöppfer, W. & Grahl, B. (2009). Ökobilanz (LCA): Ein Leitfaden für Ausbildung und Beruf. Weinheim: Wiley.
- KPMG International. (2014). A new vision of value: Connecting corporate and societal value creation. Amstelveen, Netherlands: KPMG International.
- KPMG (2018) How to report on the SDGs – What it looks like and why it matters. Amstelveen, Netherlands: KPMG International.
- Kroeger, A. & Weber, C. (2015). Developing a conceptual framework for comparing social value creation. Academy of Management Review, 40(1), 43–70.
- Kühnen, M., Hahn, R., Silva, S. L. & Schaltegger, S. (2017). Verständnis und Messung sozialer und positiver Nachhaltigkeitswirkungen: Erkenntnisse aus Literatur, Praxis und Delphi-Studien – Arbeitspapier Nr. 1 im Rahmen des Projekts „Der Handabdruck: Ein komplementäres Maß positiver Nachhaltigkeitswirkung von Produkten“ (in Zusammenarbeit mit Eberle, U., Schmid, M., Beckmann, J., Eisenhauer, P. & Hermann, C.). Lüneburg: Verein CSM e.V.
- Lindner, JP; Eberle, U.; Schmincke, E.; Luick, R.; Niblick, B.; Brethauer, L.; Knüpfper, E.; Beck, T.; Schwendt, P.; Schestak, I.; Arana, D. (in Vorbereitung): Biodiversität in Ökobilanzen. Abschlussbericht des des F+E-Vorhabens "Weiterentwicklung der Ökobilanzen durch Integration der Biodiversitätsauswirkun-gen von Produkten" (FKZ 3511 82 3100); wird veröffentlicht in der Reihe BfN-Schriften.
- Linstone, H. A., Turoff, M., & Helmer, O. (Eds.). (2002). The Delphi method: Techniques and applications. Online edition of the original published in 1975, Reading, MA: Addison-Wesley.

- Living Planet Index: <http://www.livingplanetindex.org/home/index> (letzter Zugriff: 26.04.2017)
- Lützkendorf, T. & Lorenz, D. P. (2005). Sustainable property investment: Valuing sustainable buildings through property performance assessment. *Building Research & Information*, 33(3), 212–234.
- Lützkendorf, T. & Lorenz, D. P. (2006). Using an integrated performance approach in building assessment tools. *Building Research & Information*, 34(4), 334–356.
- Luthans, F. (2002). Positive organizational behavior: Developing and managing psychological strengths. *Academy of Management Executive*, 16(1), 57–72.
- Lyneis, J. & Sterman, J. (2016). How to save a leaky ship: Capability traps and the failure of win-win investments in sustainability and social responsibility. *Academy of Management Discoveries*, 2(1), 7–32.
- Maas, K., Schaltegger, S. & Crutzen, N. (2016a). Reprint of advancing the integration of corporate sustainability measurement, management and reporting. *Journal of Cleaner Production*, 136, 1–4.
- Maas, K., Schaltegger, S. & Crutzen, N. (2016b). Integrating corporate sustainability assessment, management accounting, control, and reporting. *Journal of Cleaner Production*, 136, 237–248.
- Martínez-Blanco, J., Lehmann, A., Chang, Y.-J. & Finkbeiner, M. (2015). Social organizational LCA (SOLCA): A new approach for implementing social LCA. *International Journal of Life Cycle Assessment*, 20(11), 1586–1599.
- Michelsen, O. (2007). Assessment of Land Use Impact on Biodiversity. Proposal of a new methodology exemplified with forestry operations in Norway. *Int J LCA* 13 (1) 22–31.
- Milà i Canals et al. (2007): Milà i Canals, L., Bauer, C., Depestele, J., Dubruil, A., Knuchel, R.F., Gaillard, G., Michelsen, O., Müller-Wenk, R., Rydgren, B. Key Elements in a Framework for Land Use Impact Assessment Within LCA. *Int J LCA* 12 (1) 5–15.
- Müller, A.-L. & Pfleger, R. (2014). Business transformation towards sustainability. *Business Research*, 7(2), 313–350.
- Nager IT e.V.: Erklärungen zu fairem Lötzinn. Bichl <https://www.nager-it.de/static/pdf/Fairzinn.pdf>
- Nilsson, K.; Flysjö, A.; Davis, J.; Sim, S.; Unger, N.; Bell, S. (2010). Comparative life cycle assessment of margarine and butter consumed in the UK, Germany and France. *Int J LCA* 15 (9) 916–926.
- Norris, G. A. (2006). Social impacts in product life cycles: Towards life cycle attribute assessment. *International Journal of Life Cycle Assessment*, 11(S1), 97–104.
- O'Brien, M., Doig, A. & Clift, R. (1996). Social and environmental life cycle assessment (SELCA). *International Journal of Life Cycle Assessment*, 1(4), 231–237.
- OECD Green Growth: Green growth and sustainable development <http://www.oecd.org/greengrowth/> (letzter Zugriff: 26.04.2017)

- Odrich, P. (2014). Bergbauunternehmen stürzen sich auf arsenverseuchte Kupferquellen. Ingenieur.de: <https://www.ingenieur.de/technik/fachbereiche/rohstoffe/bergbauunternehmen-stuerzen-arsenverseuchte-kupferquellen/> (letzter Zugriff: 05.12.18)
- Okoli, C. & Pawlowski, S. D. (2004). The Delphi method as a research tool: An example, design considerations and applications. *Information & Management*, 42(1), 15–29.
- Ottawa-Charta (1986): Download unter http://www.euro.who.int/_data/assets/pdf_file/0006/129534/Ottawa_Charter_G.pdf (Stand: 20.03.2017)
- Oum, T. H., Pathomsiri, S. & Yoshida, Y. (2013). Limitations of DEA-based approach and alternative methods in the measurement and comparison of social efficiency across firms in different transport modes: An empirical study in Japan. *Transportation Research Part E: Logistics and Transportation Review*, 57, 16–26.
- Pauw, I. C., Kandachar, P. & Karana, E. (2014). Assessing sustainability in nature-inspired design. *International Journal of Sustainable Engineering*, 8(1), 5–13.
- Planetary Boundaries: Stockholm Resilience Centre. <http://www.stockholmresilience.org/research/planetary-boundaries.html> (letzter Zugriff: 26.04.2017)
- PricewaterhouseCoopers. (2013). Measuring and managing total impact: A new language for business decisions. London, UK: PricewaterhouseCoopers.
- Pryshlakivsky, J. & Searcy, C. (2013). Fifteen years of ISO 14040: A review. *Journal of Cleaner Production*, 57, 115–123.
- Reap, J., Roman, F., Duncan, S. & Bras, B. (2008a). A survey of unresolved problems in life cycle assessment – Part 1: Goal and scope and inventory analysis. *International Journal of Life Cycle Assessment*, 13(4), 290–300.
- Reap, J., Roman, F., Duncan, S. & Bras, B. (2008b). A survey of unresolved problems in life cycle assessment – Part 2: Impact assessment and interpretation. *The International Journal of Life Cycle Assessment*, 13(5), 374–388.
- Richard, P. J., Devinney, T. M., Yip, G. S. & Johnson, G. (2009). Measuring organizational performance: Towards methodological best practice. *Journal of Management*, 35(3), 718–804.
- Roberts Enterprise Development Fund. (2000). Social return on investment: Exploring aspects of value creation in the nonprofit sector. San Francisco, US: Roberts Enterprise Development Fund.
- Roberts Enterprise Development Fund. (2008). REDF's current approach to SROI. San Francisco, US: Roberts Enterprise Development Fund.
- Rosenbaum et al. (2008). USEtox—the UNEP-SETAC toxicity model: recommended characterisation factors for human toxicity and freshwater ecotoxicity in life cycle impact assessment. *Int J LCA First Online*, November 2008
- Sala, S., Farioli, F. & Zamagni, A. (2013). Progress in sustainability science: Lessons learnt from current methodologies for sustainability assessment: Part 1. *International Journal of Life Cycle Assessment*, 18(9), 1653–1672.

- Salvado, M., Azevedo, S., Matias, J. & Ferreira, L. (2015). Proposal of a sustainability index for the automotive industry. *Sustainability*, 7(2), 2113–2144.
- Schaltegger, S. & Burritt, R. L. (2010). Sustainability accounting for companies: Catchphrase or decision support for business leaders? *Journal of World Business*, 45(4), 375–384.
- Schaltegger, S. & Burritt, R. L. (2012). Measuring the (un-)sustainability of industrial biomass production and use. *Sustainability Accounting, Management and Policy Journal*, 3(2), 109–133.
- Schaltegger, S. & Csutora, M. (2012). Carbon accounting for sustainability and management. Status quo and challenges. *Journal of Cleaner Production*, 36, 1–16.
- Schaltegger, S. & Burritt, R. (2014). Measuring and managing sustainability performance of supply chains: Review and sustainability supply chain management framework. *Supply Chain Management: An International Journal*, 19(3), 232–241.
- Schmidt, R. C. (1997). Managing Delphi Surveys using nonparametric statistical techniques. *Decision Sciences*, 28(3), 763–774.
- Schmidt, I., M. Meurer, P. Saling, A. Kicherer, W. Reuter & C.-O. Gensch. (2004). SEEbalance: Managing sustainability of products and processes with the socio-eco- efficiency analysis by BASF. *Greener Management International* (45), 79–94.
- Schnurr, B. (2018). Masterarbeit: Analyse der Übertragbarkeit von methodischen Bestandteilen der Ökobilanzierung auf den Handabdruck von Produkten; einzusehen am Lehrstuhl Energiesysteme und Energiewirtschaft der Ruhr-Universität Bochum (<http://www.lee.ruhr-uni-bochum.de/>)
- Smirnova, J. (2015). Ist sie zu kalt, bist Du zu schwach. WELT: <https://www.welt.de/politik/ausland/article150373411/Ist-sie-zu-kalt-bist-Du-zu-schwach.html> (letzter Zugriff: 05.12.18)
- Seuring, S. (2013). A review of modeling approaches for sustainable supply chain management. *Decision Support Systems*, 54(4), 1513–1520.
- Seuring, S., Koplin, J., Behrens, T. & Schneidewind, U. (2003). Sustainability assessment in the german detergent industry: From stakeholder involvement to sustainability indicators. *Sustainable Development*, 11(4), 199–212.
- Social Enterprise London. (2013). Social impact measurement for local economies (SIMPLE). London, UK: Social Enterprise London.
- Social Value UK. (2012). A guide to social return on investment. Liverpool: Social Value UK.
- Streicher-Porte, M., Marthaler, C., Böni, H., Schluep, M., Camacho, A. & Hilty, L. M. (2009). One laptop per child, local refurbishment or overseas donations? Sustainability assessment of computer supply scenarios for schools in Colombia. *Journal of Environmental Management*, 90(11), 3498–3511.
- Tranfield, D., Denyer, D. & Smart, P. (2003). Towards a methodology for developing evidenceinformed management knowledge by means of systematic review. *British Journal of Management*, 14(3), 207–222.

- True Impact LLC. (2017). ROI Tracker. URL: <http://www.trueimpact.com/ROI-analysis-training> (zuletzt geprüft am 2. Februar 2017).
- UN. (2015). Sustainable development goals: 17 goals to transform our world. URL: <http://www.un.org/sustainabledevelopment/sustainable-development-goals/> (zuletzt geprüft am 2. Februar 2017).
- UN. (2015): Sustainable Development Knowledge Platform: <https://sustainabledevelopment.un.org> (Stand 29.03.2017)
- UNEP & SETAC. (2009). Guidelines for social life cycle assessment of products. Paris: UNEP. UNEP & SETAC. (2013). The methodological sheets for subcategories in social life cycle assessment (S-LCA). Paris: UNEP.
- Vereinte Nationen (2015). Transformation unserer Welt: Die Agenda 2030 für nachhaltige Entwicklung, Download unter <http://www.un.org/depts/german/gv-70/a70-l1.pdf> (Stand 29.03.17)
- Vinyes, E., Oliver-Solà, J., Ugaya, C., Rieradevall, J. & Gasol, C. M. (2013). Application of LCSA to used cooking oil waste management. *International Journal of Life Cycle Assessment*, 18(2), 445–455.
- Vision 2050: <http://www.wbcsd.org/Overview/About-us/Vision-2050> (letzter Zugriff: 26.04.2017)
- Weidema, B. (2014). Has ISO 14040/44 failed its role as a standard for life cycle assessment? *Journal of Industrial Ecology*, 18(3), 324–326.
- Wilhelm, M., Hutchins, M., Mars, C. & Benoit-Norris, C. (2015). An overview of social impacts and their corresponding improvement implications: A mobile phone case study. *Journal of Cleaner Production*, 102, 302–315.
- William Davidson Institute. (2017). Base of the pyramid impact assessment framework. URL: <http://wdi.umich.edu/initiatives/performance-measurement/> (zuletzt geprüft am 2. Februar 2017).
- WWF (2006): World Wildlife Fund. 2006. WildFinder: Online database of species distributions, ver. Jan-06. www.worldwildlife.org/WildFinder
- Zamagni, A., Masoni, P., Buttol, P., Raggi, A. & Buonamici, R. (2012). Finding life cycle assessment research direction with the aid of meta-analysis. *Journal of Industrial Ecology*, 16, S39- S52.
- Zellmer-Bruhn, M. & Gibson, C. (2006). Multinational organization context: Implications for team learning and performance. *Academy of Management Journal*, 49(3), 501-518.